Memory Semis Monthly

July '26 Edition: Further pricing upside amidst LTA negotiations

Equities

Global

Semiconductors

Nicolas Gaudois
Analyst
nicolas.gaudois@ubs.com
+65-6495 5148

Timothy Arcuri
Analyst
timothy.arcuri@ubs.com
+1-415-352 5676

Kenji Yasui
Analyst
kenji.yasui@ubs.com
+81-3-5208 6211

Jimmy Yoon
Analyst
jimmy.yoon@ubs.com
+65-6495 4617

Gianmarco Vella
Associate Analyst
gianmarco.vella@ubs.com
+1-415-352 4555

Atsuhiro Kinoshita
Analyst
atsuhiro.kinoshita@ubs.com
+81-3-5208 6768

Sunny Lin
Analyst
sunny.lin@ubs.com
+886-2-8722 7346

Jimmy Yu
Analyst
S1460517080002
jimmy.yu@ubs.com
+86-21-3866 8880

Randy Abrams
Analyst
randy.abrams@ubs.com
+886-2-8722 7338

Francois-Xavier Bouvignies
Analyst
francois.bouvignies@ubs.com
+44-20-7568 7105

Upside to 2H26 memory pricing

Based upon our industry checks, we increase base-line DDR contract pricing to +32% QoQ in 3Q26 (was +17%) and +18% QoQ in 4Q26 (was +12%), after +67% QoQ in 2Q26. Depending upon pricing under longer term agreements (LTAs), as well as HBM pricing and mix, the QoQ ASP comparisons will vary though by memory vendor. We continue to forecast the DRAM industry to be undersupplied until at least 2Q28. The gap between bit demand growth in 2027 (+36.2% YoY) and supply (+19.3%) is too wide to close by then. If we assume no inventory digestion downstream in 2027 (i.e., no customers' inventory rebuild in 2H26), then the sufficiency ratio would actually deteriorate to -13.6% from -8.1% in 2026E. Both are unprecedented levels over the last 30 years. As for NAND flash, we now forecast +30% QoQ in 3Q26 (was +17%) and +12% QoQ (unchanged) in 4Q26 after +67% QoQ also in 2Q. We forecast the NAND upcycle to continue til at least 4Q27. Net net, this leaves us forecasting memory industry revenues to reach US$992bn in 2026 and US$1,763bn in 2027. The main risk to this unprecedented upcycle remains affordability by customers, notably hyperscalers which will have to continue to tap capital markets to finance this capex.

LTA negotiations continue

Revised LTA negotiations, which incorporate more fixed terms for volumes and pricing, continue. We see some differences between Micron's SCAs, with price "floor" and "ceiling", and those of the Korean memory makers, which we believe have a portion of volumes fixed (c. 60-70% over a 5-year contract), at fixed pricing, to which a variable component (volume and pricing) is added. We believe Samsung is progressing in completing by 3Q more revised LTA agreements with another handful of large customers. The focus remains on DDR5, for which we would expect a target of 50-70% of volumes being tied up to LTAs. As for SK Hynix, we believe negotiations continue to focus on the larger hyperscale customers for both DDR5 and NAND flash.

Slightly raising HBM demand forecasts, Meta not an issue

We tweak our HBM industry demand forecast to 33.1bn Gb in 2026 (+90% YoY) and 58.7bn Gb in 2027 (+77% YoY) from 32.9/58.0bn prior. We assume HBM procurement equivalent to 8.5mil Nvidia AI GPU units in 2026E, and 11.0mil in 2027E, consistent with our recent CoWoS estimate revision (link). We revise our estimates for HBM-equivalent Google TPU units to 4.2mil for 2026E and 9.1mil (up) for 2027E, also tweaking up HBM procurement for AMD and AWS for 2027. Meanwhile, recent press reports of Meta possibly selling compute to outside customers (link) negatively impacted memory stocks earlier this week. Still, this is likely monetisation of older assets prior to rolling out further Blackwell/Rubin, as well as MTIA capacity (see also note from Steven Ju: link). As such, we see no impact on HBM procurement.

Pullback in memory semis stocks likely temporary

Memory stocks are on average down 17% from their June peak, after outperforming Semis, Tech, and GEM YTD. Some of this could be over-crowding. Yet, fundamentals remain strong, with the memory industry set to generate close to US$1.2tn of FCF in 2027E. We believe this will ultimately lead to a step-up in returns to shareholders. We remain positive on the group, with Samsung Key Call Buy (PT Won550k), and Buy ratings on SK Hynix (PT 3.20m from 3.00m), MU (PT US$1,625), Kioxia (PT ¥132,000) and Nanya Tech (PT NT$495).

This report has been prepared by UBS Securities Pte. Ltd. (Reg. No. 198500648C) **ANALYST CERTIFICATION AND REQUIRED DISCLOSURES, INCLUDING INFORMATION ON THE QUANTITATIVE RESEARCH REVIEW PUBLISHED BY UBS, BEGIN ON PAGE 16.**

KEY CHARTS

Figure 1: AI server GPU/accelerator unit forecasts ('000 units)

	HBM Gen.	HBM Content (GB)			2025	1Q26	2Q26E	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E
NVIDIA		**2025E**	**2026E**	**2027E**											
L40S	GDDR6	48			38.1										
H200	HBM3E	141			592.7										
H20 / H20E	HBM3 / 3E	96	96		441.0										
B200 (8Hi)	HBM3E	192	192		1,440.9										
GB200 (2x GPU - 8Hi)	HBM3E	384	384		1,080.4	102.9				102.9					
B300 (12Hi)	HBM3E	288	288	288	587.4	267.0	300.0	250.0	170.0	987.0	100.0				100.0
GB300 (2x GPU - 12Hi)	HBM3E	576	576	576	778.2	658.0	670.0	600.0	520.0	2,448.0	280.0	180.0	110.0	60.0	630.0
R200 (12Hi)	HBM4	288	288	288			20.0	200.0	380.0	600.0	360.0	400.0	300.0	210.0	1,270.0
RV200 (2x GPU - 12Hi)	HBM4	576	576	576			40.0	250.0	480.0	770.0	570.0	700.0	670.0	520.0	2,460.0
R300 (12Hi)	HBM4E		384	384									130.0	270.0	400.0
RV300 (4x GPU - 12Hi)	HBM4E		768	768									200.0	500.0	700.0
*Other GPU/Accel	HBM3 / 3E	288	384	576	106.9	60.0	80.0	80.0	80.0	300.0	70.0	60.0	60.0	50.0	240.0
Total Units					**5,065.6**	**1,087.9**	**1,110.0**	**1,380.0**	**1,630.0**	**5,207.9**	**1,380.0**	**1,340.0**	**1,470.0**	**1,610.0**	**5,800.0**
% YoY					*3.2%*	*-17.2%*	*0.0%*	*8.2%*	*19.4%*	*2.8%*	*26.9%*	*20.7%*	*6.5%*	*-1.2%*	*11.4%*
% QoQ						*-20.3%*	*2.0%*	*24.3%*	*18.1%*		*-15.3%*	*-2.9%*	*9.7%*	*9.5%*	
Total GPUs					**6,924.3**	**1,848.8**	**1,820.0**	**2,230.0**	**2,630.0**	**8,528.8**	**2,230.0**	**2,220.0**	**2,850.0**	**3,690.0**	**10,990.0**
% YoY					*37.9%*	*27.2%*	*22.9%*	*19.6%*	*23.7%*	*23.2%*	*20.6%*	*22.0%*	*27.8%*	*40.3%*	*28.9%*
% QoQ						*-13.0%*	*-1.6%*	*22.5%*	*17.9%*		*-15.2%*	*-0.4%*	*28.4%*	*29.5%*	
AMD															
MI300	HBM3	192	192		71.3										
MI325X (12Hi)	HBM3E	256	256		223.8	18.0				18.0					
MI355X (12Hi)	HBM3E	288	288	288	220.7	170.0	150.0	80.0	70.0	470.0					
MI308X	HBM3	192	192	192	95.8	29.0	28.0	5.0		62.0					
MI450 (12Hi)	HBM4		432	432				45.0	230.0	275.0	350.0	450.0	470.0	380.0	1,650.0
MI5XX (16Hi)	HBM4E			1024									60.0	160.0	220.0
*Other GPU/Accel	HBM3	192	192	288	4.2	1.4	1.5	1.5	1.5	5.8	2.0	2.0	2.0	2.0	8.0
Total Units					**615.8**	**218.4**	**179.5**	**131.5**	**301.5**	**830.8**	**352.0**	**452.0**	**532.0**	**542.0**	**1,878.0**
% YoY					*41.1%*	*81.0%*	*86.4%*	*-26.5%*	*37.0%*	*34.9%*	*61.2%*	*151.8%*	*304.7%*	*79.8%*	*126.0%*
% QoQ						*-0.7%*	*-17.8%*	*-26.8%*	*129.3%*		*16.8%*	*28.4%*	*17.7%*	*1.9%*	
Google															
TPU v5e	HBM2E	32	32		134										
TPU v5p	HBM2E	96	96		1,916										
TPU v6e	HBM3E	32	32	32	426	205	224	141	70	640					
TPU v7	HBM3E	192	192	192	90	520	720	800	750	2,790	640	420	300	180	1,540
TPU v8i (Dual die AX)	HBM3E		288	288				25	225	250	740	940	980	840	3,500
TPU v8t (Single die X)	HBM3E		216	216				70	380	450	850	1,040	1,160	950	4,000
*Other TPU/Accel	HBM3	96	144	144	77	7	9	10	10	37	15	15	15	15	60
Total Units					**2,642.5**	**732.0**	**953.4**	**1,046.2**	**1,435.4**	**4,167.0**	**2,245.0**	**2,415.0**	**2,455.0**	**1,985.0**	**9,100.0**
% YoY					*18.6%*	*14.3%*	*66.1%*	*82.4%*	*68.0%*	*57.7%*	*206.7%*	*153.3%*	*134.7%*	*38.3%*	*118.4%*
% QoQ						*-14.3%*	*30.2%*	*9.7%*	*37.2%*		*56.4%*	*7.6%*	*1.7%*	*-19.1%*	
Amazon															
Inferentia 2	HBM2E	32	32		88										
Trainium 2 / 2.5	HBM3E	96	144	144	496	130	220	140	100	590	50	10			60
Inferentia 3	HBM3	96	96	96	162	30	28	20	18	96	15	10			25
Trainium 3	HBM3E	144	144	144			320	620	840	1,780	840	780	640	540	2,800
Trainium 4	HBM4		256	256									40	240	280
*Other Accel	HBM3	144	144	144	37	5	6	8	10	28	8	8	7	7	30
Total Units					**783.2**	**164.8**	**573.7**	**787.8**	**967.6**	**2,493.9**	**913.0**	**808.0**	**687.0**	**787.0**	**3,195.0**
% YoY					*-7.7%*	*-16.5%*	*179.8%*	*301.7%*	*423.9%*	*218.4%*	*454.0%*	*40.8%*	*-12.8%*	*-18.7%*	*28.1%*
% QoQ						*-10.8%*	*248.1%*	*37.3%*	*22.8%*		*-5.6%*	*-11.5%*	*-15.0%*	*14.6%*	
Intel / Habana															
Gaudi 3	HBM2 / 3E	288	288	288	188	30	18	8	0	56					
Gaudi 4	HBM3E	288	288	288				10	25	35	25	20	15	10	70
*Others	HBM2 / 3E	192	192	192	25	6	4	4	5	18	4	4	4	4	16
Total Units					**213.2**	**36.0**	**21.6**	**21.6**	**30.0**	**109.2**	**29.0**	**24.0**	**19.0**	**14.0**	**86.0**
% YoY					*-32.1%*	*-20.3%*	*-60.1%*	*-62.7%*	*-46.3%*	*-48.8%*	*-19.4%*	*11.1%*	*-12.0%*	*-53.3%*	*-21.2%*
% QoQ						*-35.6%*	*-40.0%*	*0.0%*	*38.9%*		*-3.3%*	*-17.2%*	*-20.8%*	*-26.3%*	
Microsoft															
Maia 100	HBM3	120	120	120	34	3	1			4					
Maia 200	HBM3E		288	288		2	6	10	14	33	14	10	7	4	35
Maia 300	HBM4			288									8	14	22
*Others	HBM3	120	192	192	8	2	2	2	2	8	2	2	2	2	8
Total Units					**41.8**	**7.3**	**8.9**	**12.5**	**16.1**	**44.7**	**16.0**	**12.0**	**17.0**	**20.0**	**65.0**
% YoY					*22.3%*	*-27.1%*	*-19.2%*	*3.9%*	*83.6%*	*7.1%*	*119.5%*	*35.0%*	*36.3%*	*24.4%*	*45.3%*
% QoQ						*-16.7%*	*21.9%*	*40.3%*	*28.9%*		*-0.4%*	*-25.0%*	*41.7%*	*17.6%*	
Meta															
MTIA v1	LPDDR5	64	64		421	20	5			25					
MTIA v2	LPDDR5	128	128		30	20	8	2		30					
MTIA v3	HBM3E	192	192	192	5	15	20	35	40	110	40	30	20	15	105
MTIA v4	HBM4		432	432							40	80	100	120	340
*Others	LPDDR5	64	128	128	28	3	1	1	1	6	5	5	5	5	20
Total Units					**483.9**	**57.8**	**34.3**	**38.1**	**41.2**	**171.4**	**85.0**	**115.0**	**125.0**	**140.0**	**465.0**
% YoY					*19.3%*	*-43.2%*	*-71.1%*	*-71.4%*	*-68.3%*	*-64.6%*	*47.2%*	*235.1%*	*228.0%*	*239.8%*	*171.3%*
% QoQ						*-55.6%*	*-40.6%*	*11.0%*	*8.1%*		*106.3%*	*35.3%*	*8.7%*	*12.0%*	
OpenAI															
Titan v1	HBM3E		216	216				20	150	170	160	170	130	100	560
Titan v2	HBM4E		576	576									50	130	180
Total Units								**20.0**	**150.0**	**170.0**	**160.0**	**170.0**	**180.0**	**230.0**	**740.0**
% YoY													*800.0%*	*53.3%*	*335.3%*
% QoQ									*650.0%*		*6.7%*	*6.3%*	*5.9%*	*27.8%*	
Tesla															
Dojo	HBM3	64	64	64	62	17	18	15	14	64	15	15	18	15	63
*Others	HBM3	64	64	64	44	7	7	6	6	26	6	6	5	5	22
Total Units					**105.9**	**23.8**	**25.2**	**21.0**	**19.6**	**89.6**	**21.0**	**21.0**	**23.0**	**20.0**	**85.0**
% YoY					*-39.1%*	*-12.4%*	*-9.3%*	*-24.1%*	*-15.9%*	*-15.4%*	*-11.8%*	*-16.7%*	*9.5%*	*2.0%*	*-5.2%*
% QoQ						*2.0%*	*5.9%*	*-16.7%*	*-6.6%*		*7.1%*	*0.0%*	*9.5%*	*-13.0%*	
Others															
Startups	HBM3	192	288	288	657	150	150	155	155	610	170	180	185	190	725
Others	HBM2E /3	192	288	288	361	30	30	39	39	138	60	63	65	67	254
Total Units					**1,018.5**	**180.0**	**180.0**	**193.8**	**193.8**	**747.5**	**229.5**	**243.0**	**249.8**	**256.5**	**978.8**
% YoY					*58.8%*	*-28.3%*	*-31.6%*	*-25.7%*	*-20.4%*	*-26.6%*	*27.5%*	*35.0%*	*28.9%*	*32.4%*	*30.9%*
% QoQ						*-26.1%*	*0.0%*	*7.6%*	*0.0%*		*18.5%*	*5.9%*	*2.8%*	*2.7%*	
Total GPU / Accelerator Units ('000)					**12,829.1**	**3,268.9**	**3,796.6**	**4,502.4**	**5,785.1**	**17,352.9**	**6,280.5**	**6,480.0**	**7,137.8**	**7,684.5**	**27,582.8**
% YoY					*27.0%*	*14.8%*	*34.1%*	*36.3%*	*50.4%*	*35.3%*	*92.1%*	*70.7%*	*58.5%*	*32.8%*	*59.0%*
% QoQ						*-15.0%*	*16.1%*	*18.6%*	*28.5%*		*8.6%*	*3.2%*	*10.2%*	*7.7%*	

Source: Company data, UBS estimates

Figure 2: HBM demand forecasts by GPU/accelerator

	HBM Gen.	HBM Content (GB)			2025	1Q26E	2Q26E	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E
NVIDIA		**2025E**	**2026E**	**2027E**											
H200	HBM3E	141			83,571										
H20 / H20E	HBM3 / 3E	96	96		42,336										
B200 (8Hi)	HBM3E	192	192		276,649										
GB200 (2x GPU - 8Hi)	HBM3E	384	384		414,872	39,504				39,504					
B300 (12Hi)	HBM3E	288	288	288	169,176	76,896	86,400	72,000	48,960	284,256	28,800				28,800
GB300 (2x GPU - 12Hi)	HBM3E	576	576	576	448,259	379,008	385,920	345,600	299,520	1,410,048	161,280	103,680	63,360	34,560	362,880
R200 (12Hi)	HBM4	288	288	288			5,760	57,600	109,440	172,800	103,680	115,200	86,400	60,480	365,760
RV200 (2x GPU - 12Hi)	HBM4	576	576	576			23,040	144,000	276,480	443,520	328,320	403,200	385,920	299,520	1,416,960
RV200 (2x GPU - 12Hi)	HBM4	576	576	576									49,920	103,680	153,600
Rubin Ultra (16Hi)	HBM4E		768	768									153,600	384,000	537,600
*Other GPU/Accel	HBM3 / 3E	288	384	576	30,787	23,040	30,720	30,720	30,720	115,200	40,320	34,560	34,560	28,800	138,240
Total ('000 GB)					**1,465,650**	**518,448**	**531,840**	**649,920**	**765,120**	**2,465,328**	**662,400**	**656,640**	**773,760**	**911,040**	**3,003,840**
Total (bn Gb)					**11.73**	**4.15**	**4.25**	**5.20**	**6.12**	**19.72**	**5.30**	**5.25**	**6.19**	**7.29**	**24.03**
AMD															
MI300	HBM3	192	192		13,685										
MI325X (12Hi)	HBM3E	256	256		57,294	4,608				4,608					
MI355X (12Hi)	HBM3E	288	288	288	63,569	48,960	43,200	23,040	20,160	135,360					
MI308X	HBM3	192	192	192	18,394	5,568	5,376	960		11,904					
MI450 (12Hi)	HBM4		432	432				19,440	99,360	118,800	151,200	194,400	203,040	164,160	712,800
MI5XX (16Hi)	HBM4E			1,024									61,440	163,840	225,280
*Other GPU/Accel	HBM3	192	192	288	812	274	285	282	281	1,123	576	576	576	576	2,304
Total ('000 GB)					**153,754**	**59,410**	**48,861**	**43,722**	**119,801**	**271,795**	**151,776**	**194,976**	**265,056**	**328,576**	**940,384**
Total (bn Gb)					**1.23**	**0.48**	**0.39**	**0.35**	**0.96**	**2.17**	**1.21**	**1.56**	**2.12**	**2.63**	**7.52**
Google															
TPU v5e	HBM2E	32	32		4,282										
TPU v5p	HBM2E	96	96		183,913										
TPU v6e	HBM3E	32	32	32	13,632	6,554	7,168	4,506	2,253	20,480					
TPU v7	HBM3E	192	192	192	17,280	99,840	138,240	153,600	144,000	535,680	122,880	80,640	57,600	34,560	295,680
TPU v8i (Dual die AX)	HBM3E		288	288				7,200	64,800	72,000	213,120	270,720	282,240	241,920	1,008,000
TPU v8t (Single die X)	HBM3E		216	216				15,120	82,080	97,200	183,600	224,640	250,560	205,200	864,000
TPU v9AX	HBM4E		768	768										0	0
TPU v9X	HBM4E		384	384										0	0
*Other TPU/Accel	HBM3	96	144	144	7,389	1,044	1,359	1,492	1,440	5,335	2,160	2,160	2,160	2,160	8,640
Total ('000 GB)					**226,495**	**107,437**	**146,767**	**181,917**	**294,573**	**730,695**	**521,760**	**578,160**	**592,560**	**483,840**	**2,176,320**
Total (bn Gb)					**1.81**	**0.86**	**1.17**	**1.46**	**2.36**	**5.85**	**4.17**	**4.63**	**4.74**	**3.87**	**17.41**
Amazon															
Inferentia 2	HBM2E	32	32		2,823										
Trainium 2 / 2.5	HBM3E	96	144	144	47,574	18,720	31,680	20,160	14,400	84,960	7,200	1,440			8,640
Inferentia 3	HBM3	96	96	96	15,563	2,880	2,688	1,920	1,728	9,216	1,440	960			2,400
Trainium 3	HBM3E	144	144	144			46,080	89,280	120,960	256,320	120,960	112,320	92,160	77,760	403,200
Trainium 4	HBM4		256	256									10,240	61,440	71,680
*Other Accel	HBM3	144	144	144	5,375	691	818	1,123	1,380	4,012	1,152	1,152	1,008	1,008	4,320
Total ('000 GB)					**71,336**	**22,291**	**81,266**	**112,483**	**138,468**	**354,508**	**130,752**	**115,872**	**103,408**	**140,208**	**490,240**
Total (bn Gb)					**0.57**	**0.18**	**0.65**	**0.90**	**1.11**	**2.84**	**1.05**	**0.93**	**0.83**	**1.12**	**3.92**
Intel															
Gaudi 3	HBM2 / 3E	288	288	288	54,212	8,640	5,184	2,304		16,128					
Gaudi 4	HBM3E	288	288	288				2,880	7,200	10,080	7,200	5,760	4,320	2,880	20,160
*Others	HBM2 / 3E	192	192	192	4,789	1,152	691	691	960	3,494	768	768	768	768	3,072
Total ('000 GB)					**59,001**	**9,792**	**5,875**	**5,875**	**8,160**	**29,702**	**7,968**	**6,528**	**5,088**	**3,648**	**23,232**
Total (bn Gb)					**0.47**	**0.08**	**0.05**	**0.05**	**0.07**	**0.24**	**0.06**	**0.05**	**0.04**	**0.03**	**0.19**
Microsoft															
Maia 100	HBM3	120	120	120	4,050	360	120			480					
Maia 200	HBM3E		288	288		660	1,696	3,016	4,052	9,424	4,032	2,880	2,016	1,152	10,080
Maia 300	HBM4			288									2,304	4,032	6,336
*Others	HBM3	120	192	192	960	384	384	384	384	1,536	384	384	384	384	1,536
Total ('000 GB)					**5,010**	**1,404**	**2,200**	**3,400**	**4,436**	**11,440**	**4,416**	**3,264**	**4,704**	**5,568**	**17,952**
Total (bn Gb)					**0.04**	**0.01**	**0.02**	**0.03**	**0.04**	**0.09**	**0.04**	**0.03**	**0.04**	**0.04**	**0.14**
Meta															
MTIA v3	HBM3E	192	192	192	960	2,880	3,840	6,720	7,680	21,120	7,680	5,760	3,840	2,880	20,160
MTIA v4	HBM4		432	432							17,280	34,560	43,200	51,840	146,880
Total ('000 GB)					**960**	**2,880**	**3,840**	**6,720**	**7,680**	**21,120**	**24,960**	**40,320**	**47,040**	**54,720**	**167,040**
Total (bn Gb)					**0.01**	**0.02**	**0.03**	**0.05**	**0.06**	**0.17**	**0.20**	**0.32**	**0.38**	**0.44**	**1.34**
OpenAI															
Titan v1	HBM3E		216	216				4,320	32,400	36,720	34,560	36,720	28,080	21,600	120,960
Titan v2	HBM4E		576	576									28,800	74,880	103,680
Total ('000 GB)								**4,320**	**32,400**	**36,720**	**34,560**	**36,720**	**56,880**	**96,480**	**224,640**
Total (bn Gb)								**0.03**	**0.26**	**0.29**	**0.28**	**0.29**	**0.46**	**0.77**	**1.80**
Others															
Tesla	HBM3	64	64		6,780	1,523	1,613	1,344	1,255	5,735	1,344	1,344	1,472	1,280	5,440
Startups / Etc.	HBM2E / 3	192	288		195,559	51,840	51,840	55,800	55,800	215,280	66,096	69,984	71,928	73,872	281,880
Total ('000 GB)					**202,339**	**53,363**	**53,453**	**57,144**	**57,055**	**221,015**	**67,440**	**71,328**	**73,400**	**75,152**	**287,320**
Total (bn Gb)					**1.62**	**0.43**	**0.43**	**0.46**	**0.46**	**1.77**	**0.54**	**0.57**	**0.59**	**0.60**	**2.30**
Total HBM Demand ('000 GB)					**2,184,545**	**775,025**	**874,103**	**1,065,502**	**1,427,694**	**4,142,324**	**1,606,032**	**1,703,808**	**1,921,896**	**2,099,232**	**7,330,968**
Total HBM Demand (bn Gb)					**17.48**	**6.20**	**6.99**	**8.52**	**11.42**	**33.14**	**12.85**	**13.63**	**15.38**	**16.79**	**58.65**
% YoY					*103.0%*	*96.5%*	*103.1%*	*76.1%*	*89.4%*	*89.6%*	*107.2%*	*94.9%*	*80.4%*	*47.0%*	*77.0%*
% QoQ						*2.8%*	*12.8%*	*21.9%*	*34.0%*		*12.5%*	*6.1%*	*12.8%*	*9.2%*	

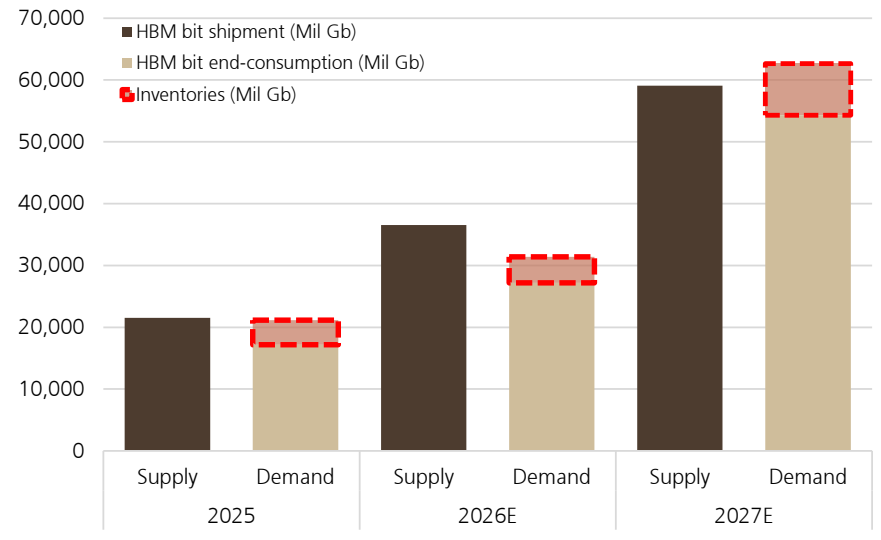
Source: Company data, UBS estimates

Figure 3: HBM vendors' bit shipments

(M 1Gb)		2025	Q126	Q226E	Q326E	Q426E	2026E	Q127E	Q227E	Q327E	Q427E	2027E
SK Hynix		**12,628**	**3,803**	**4,297**	**4,654**	**4,958**	**17,711**	**5,041**	**5,336**	**6,002**	**6,705**	**23,084**
	% YoY	*86%*	*31%*	*36%*	*44%*	*49%*	*40%*	*33%*	*24%*	*29%*	*35%*	*30%*
	% QoQ		*15%*	*13%*	*8%*	*7%*		*2%*	*6%*	*12%*	*12%*	
Samsung		**4,327**	**1,519**	**1,949**	**3,028**	**4,537**	**11,034**	**4,825**	**5,251**	**6,592**	**7,372**	**24,039**
	% YoY	*-15%*	*161%*	*155%*	*115%*	*189%*	*155%*	*218%*	*169%*	*118%*	*62%*	*118%*
	% QoQ		*-3%*	*28%*	*55%*	*50%*		*6%*	*9%*	*26%*	*12%*	
Micron		**4,584**	**1,612**	**1,821**	**2,040**	**2,305**	**7,778**	**2,536**	**2,840**	**3,152**	**3,436**	**11,964**
	% YoY	*417%*	*106%*	*60%*	*62%*	*64%*	*70%*	*57%*	*56%*	*55%*	*49%*	*54%*
	% QoQ		*15%*	*13%*	*12%*	*13%*		*10%*	*12%*	*11%*	*9%*	
Total Supply		**21,539**	**6,934**	**8,068**	**9,722**	**11,800**	**36,523**	**12,402**	**13,427**	**15,746**	**17,512**	**59,087**
	% YoY	*69%*	*62%*	*59%*	*64%*	*88%*	*70%*	*79%*	*66%*	*62%*	*48%*	*62%*
	% QoQ		*10%*	*16%*	*21%*	*21%*		*5%*	*8%*	*17%*	*11%*	

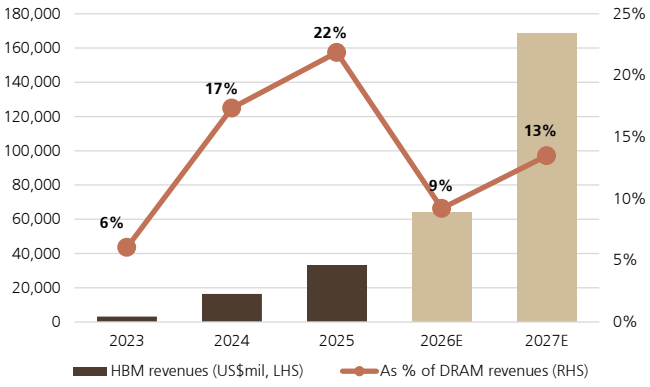
Source: Company data, UBS estimates

Figure 4: Adjusted HBM supply vs demand



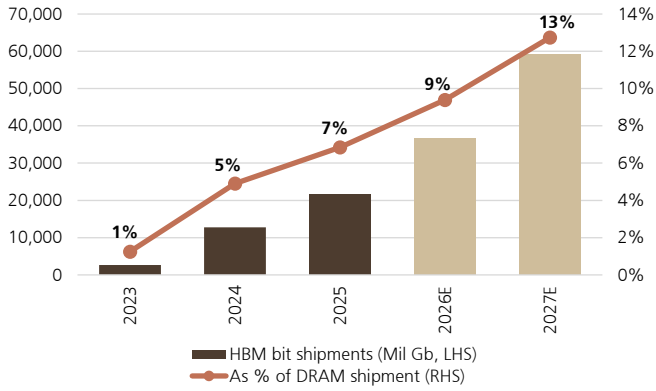
Source: Company data, UBS estimates

Figure 5: HBM as % of total DRAM revenues



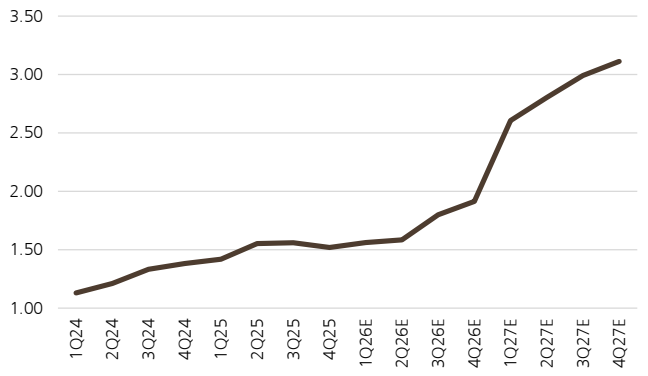
Source: UBS estimates

Figure 6: HBM as % of total DRAM bits



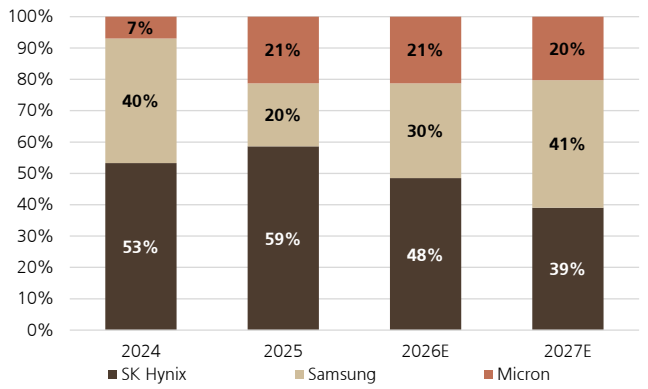
Source: UBS estimates

Figure 7: HBM ASP forecasts (US$/1Gb)



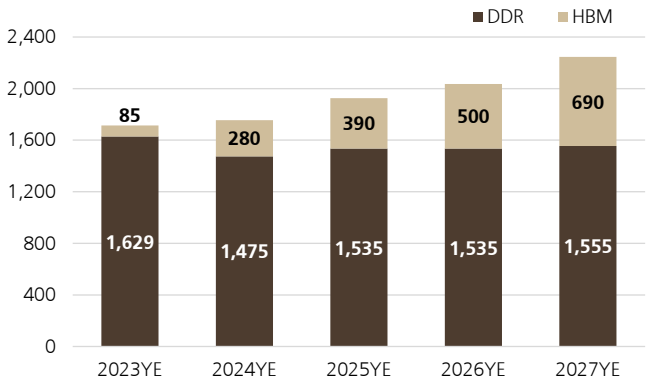
Source: UBS estimates

Figure 8: HBM bit share by vendor



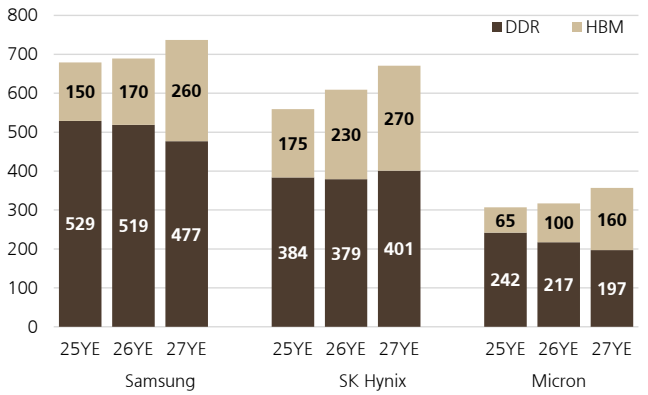
Source: Company data, UBS estimates

Figure 9: DRAM industry capacity (k wpm) breakdown between DDR and HBM



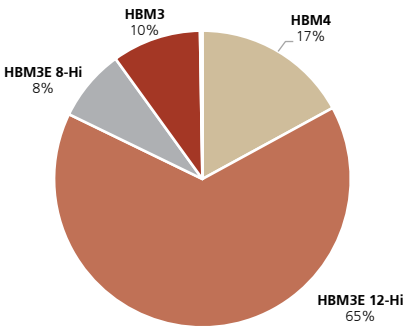
Source: Company data, UBS estimates

Figure 10: DRAM capacity (k wpm) breakdown between DDR and HBM by supplier



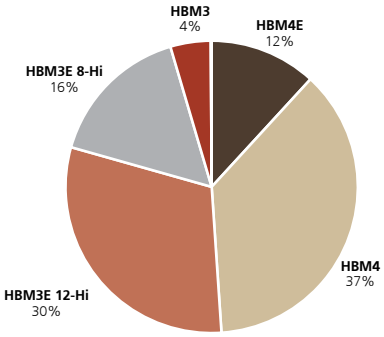
Source: Company data, UBS estimates

Figure 11: HBM demand by type (2026E)



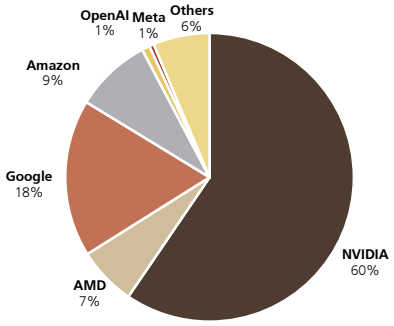
Source: UBS estimates

Figure 12: HBM demand by type (2027E)



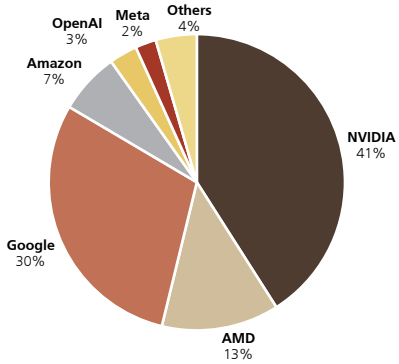
Source: UBS estimates

Figure 13: HBM demand by customer (2026E)



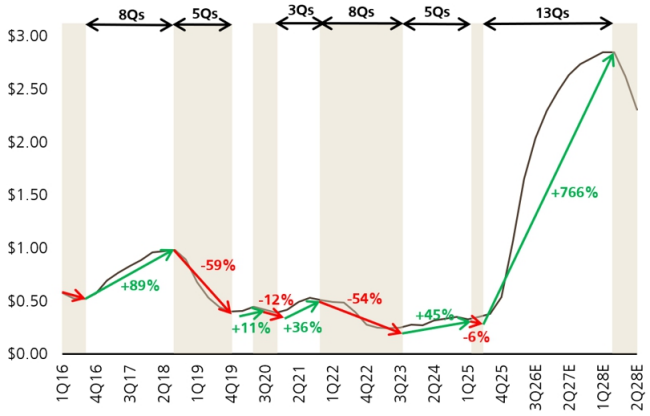
Source: UBS estimates

Figure 14: HBM demand by customer (2027E)



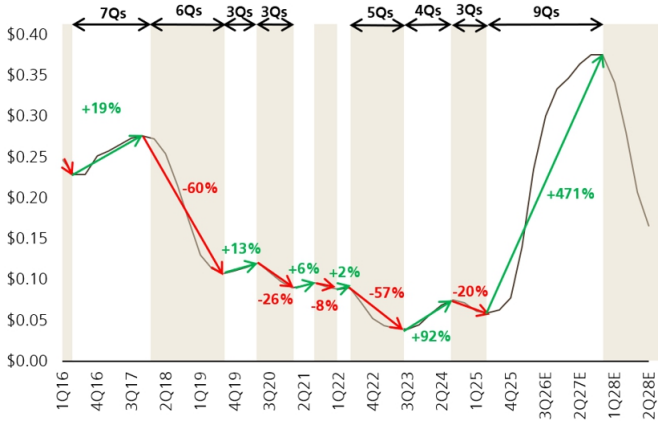
Source: UBS estimates

Figure 15: DDR cycles in context



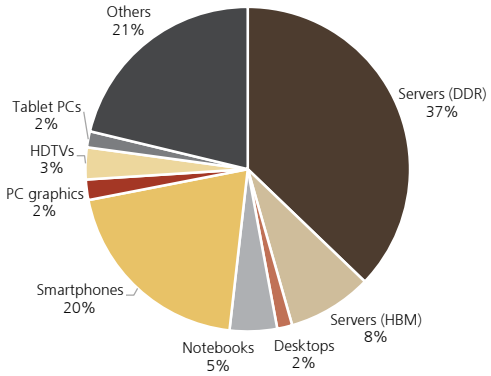
Source: Company data, UBS estimates

Figure 16: NAND cycles in context



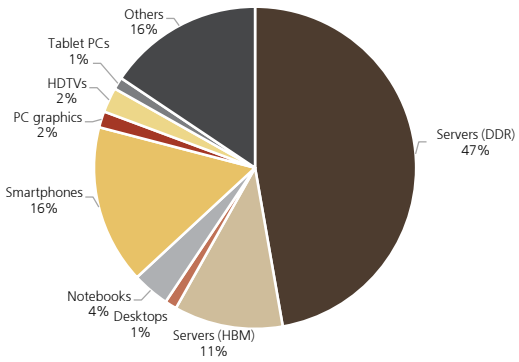
Source: Company data, UBS estimates

Figure 17: DRAM bit demand per application (2026E)



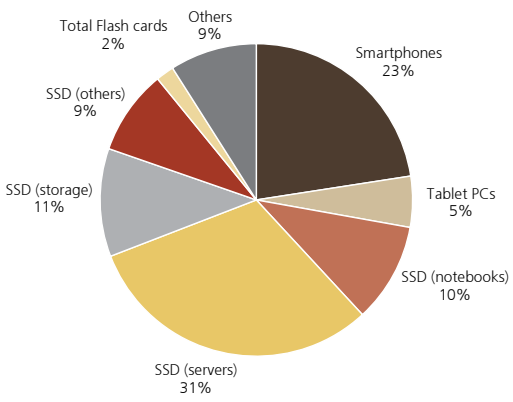
Source: Gartner, UBS estimates

Figure 18: DRAM bit demand per application (2027E)



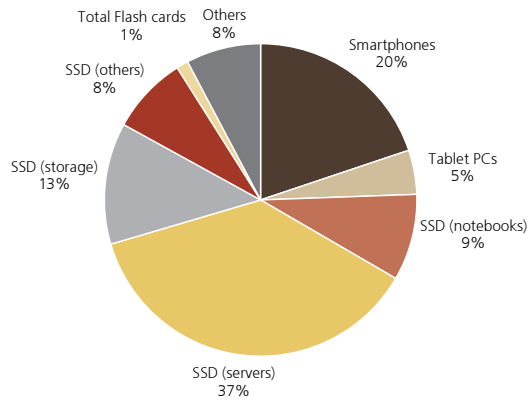
Source: Gartner, UBS estimates

Figure 19: NAND bit demand per application (2026E)



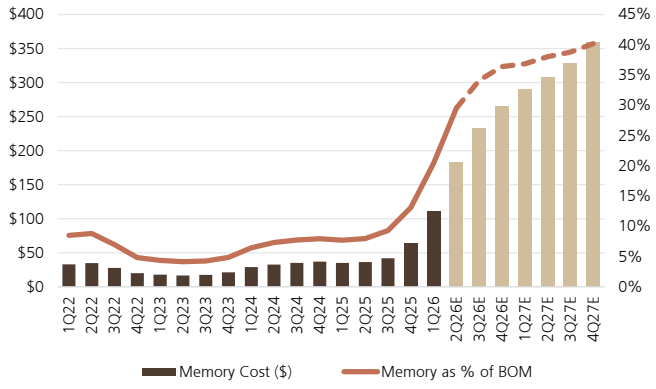
Source: Gartner, UBS estimates

Figure 20: NAND bit demand per application (2027E)



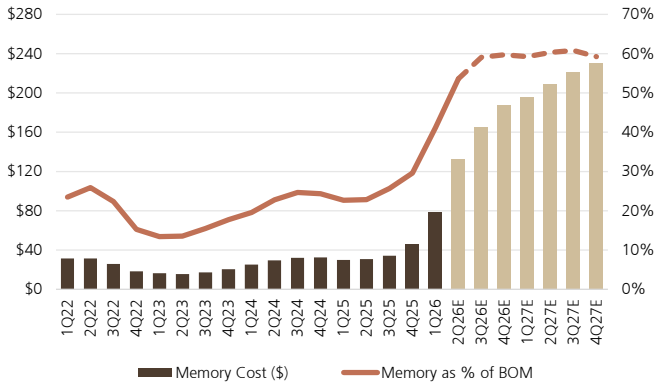
Source: Gartner, UBS estimates

Figure 21: Memory as % of flagship/premium smartphone BOM costs, 1Q22-4Q27E



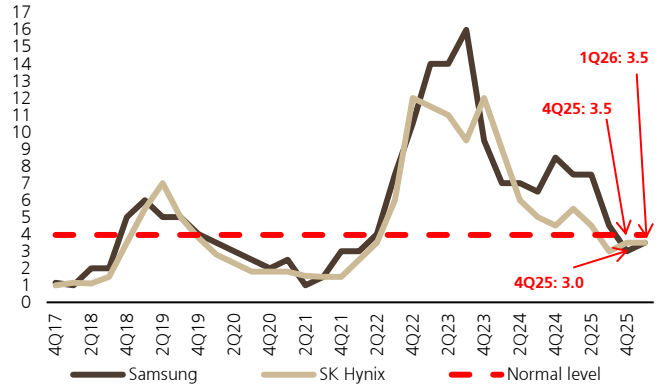
Source: Counterpoint, Fomalhaut, UBS estimates

Figure 22: Memory as % of mid-range/lower-end smartphones, 1Q22-4Q27E



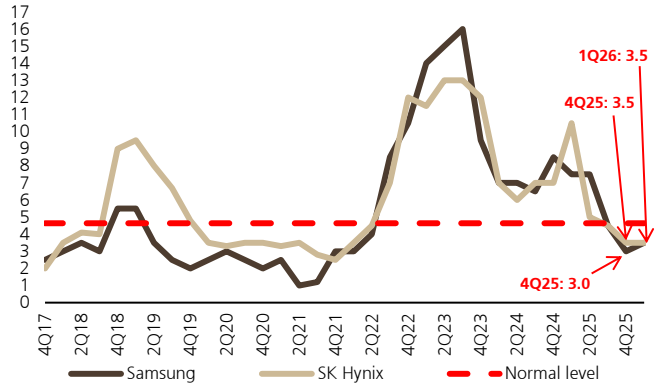
Source: UBS estimates

Figure 23: Samsung and SK Hynix – DRAM finished wafers and dies inventories



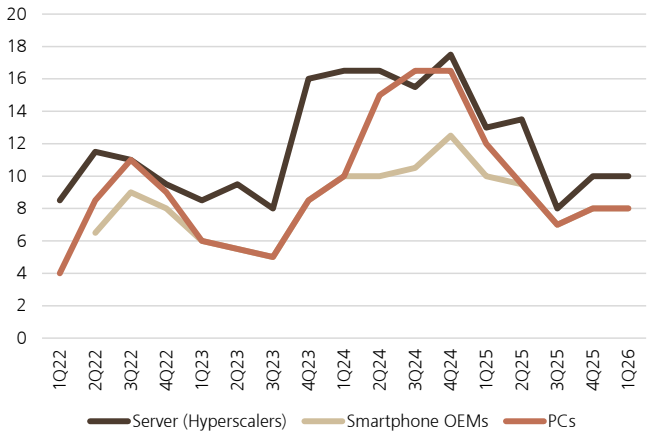
Source: Company data, UBS estimates

Figure 24: Samsung and SK Hynix – NAND finished wafers and dies inventories



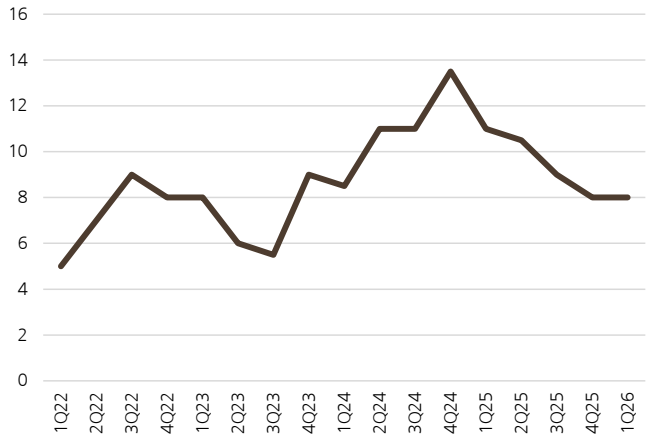
Source: Company data, UBS estimates

Figure 25: DRAM customer inventories (weeks)



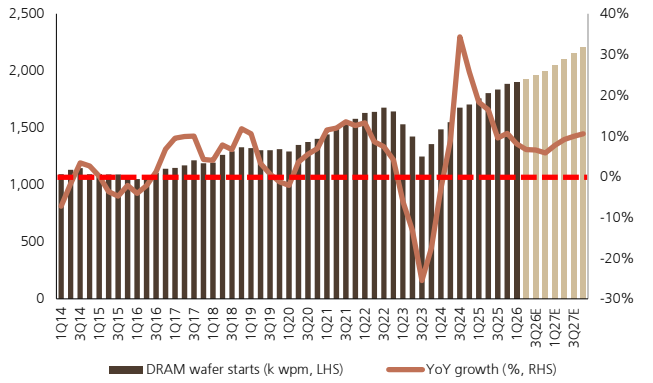
Source: Company data, UBS estimates

Figure 26: NAND customer inventories (weeks)



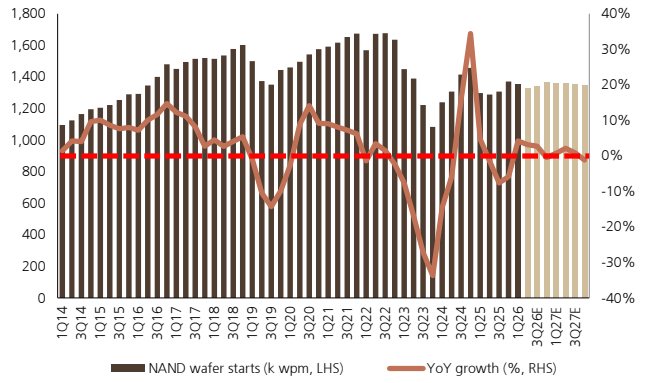
Source: Company data, UBS estimates

Figure 27: DRAM industry wafer starts and YoY growth



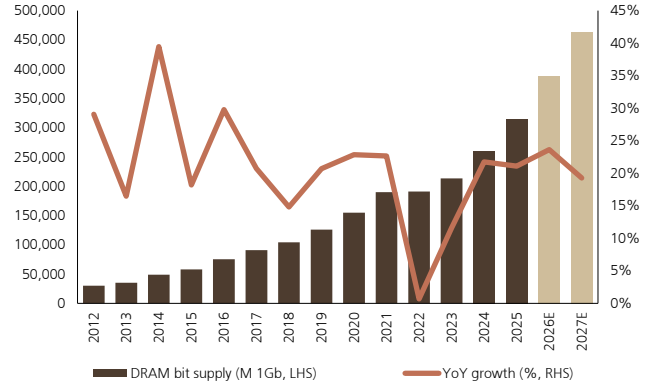
Source: Company data, UBS estimates

Figure 28: NAND industry wafer starts and YoY growth



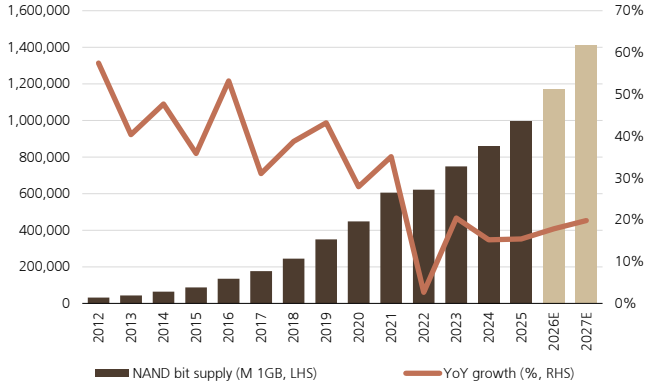
Source: Company data, UBS estimates

Figure 29: DRAM industry bit supply and YoY growth



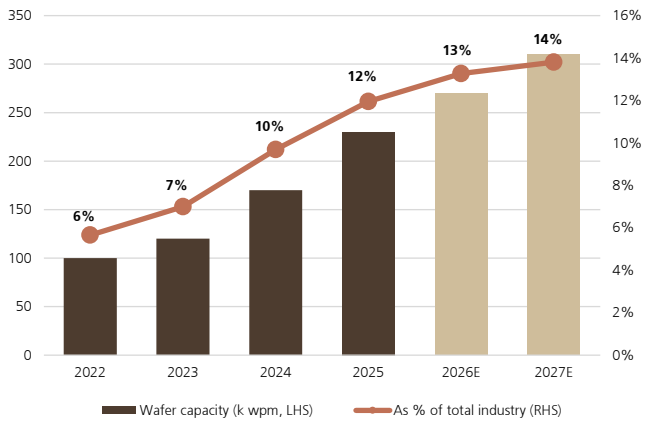
Source: Company data, UBS estimates

Figure 30: NAND industry bit supply and YoY growth



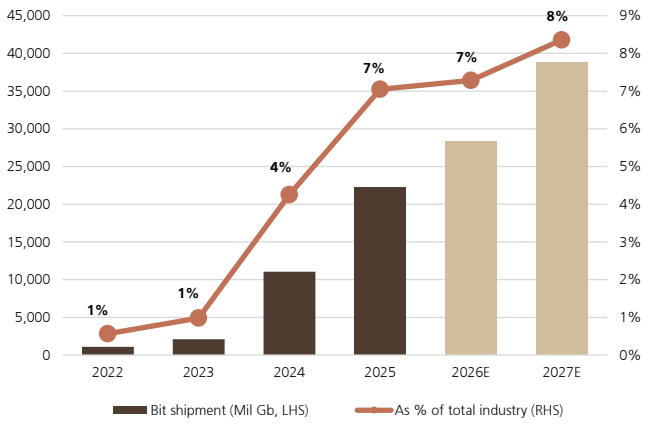
Source: Company data, UBS estimates

Figure 31: CXMT wafer capacity and as % of DRAM industry



Source: UBS estimates

Figure 32: CXMT bit shipments and as % of DRAM industry



Source: UBS estimates

Figure 33: UBS DRAM supply/demand summary

	Q125	Q225	Q325	Q425	2025	Q126	Q226E	Q326E	Q426E	2026E	Q127E	Q227E	Q327E	Q427E	2027E
Revenues ($ mil)	**26,683**	**31,996**	**39,350**	**52,865**	**150,894**	**97,020**	**153,791**	**202,404**	**238,379**	**691,595**	**265,507**	**297,562**	**332,492**	**352,577**	**1,248,138**
% QoQ	*-6.7%*	*19.9%*	*23.0%*	*34.3%*		*83.5%*	*58.5%*	*31.6%*	*17.8%*		*11.4%*	*12.1%*	*11.7%*	*6.0%*	
% YoY	*49.9%*	*41.1%*	*55.3%*	*84.9%*	*59.8%*	*263.6%*	*380.7%*	*414.4%*	*350.9%*	*358.3%*	*173.7%*	*93.5%*	*64.3%*	*47.9%*	*80.5%*
Supply (Shipments Based) (M 1Gb)	**64,968**	**76,461**	**85,829**	**87,246**	**314,505**	**88,812**	**93,489**	**100,732**	**105,801**	**388,834**	**106,695**	**112,460**	**120,097**	**124,562**	**463,814**
% QoQ	*-1.9%*	*17.7%*	*12.3%*	*1.7%*		*1.8%*	*5.3%*	*7.7%*	*5.0%*		*0.8%*	*5.4%*	*6.8%*	*3.7%*	
% YoY	*5.9%*	*15.9%*	*29.7%*	*31.7%*	*21.1%*	*36.7%*	*22.3%*	*17.4%*	*21.3%*	*23.6%*	*20.1%*	*20.3%*	*19.2%*	*17.7%*	*19.3%*
Demand (End Consumption) (M 1Gb)	**69,879**	**73,066**	**87,229**	**94,090**	**324,264**	**89,574**	**91,832**	**101,863**	**110,767**	**394,034**	**113,879**	**123,806**	**140,679**	**158,302**	**536,666**
% QoQ	*0.2%*	*4.6%*	*19.4%*	*7.9%*		*-4.8%*	*2.5%*	*10.9%*	*8.7%*		*2.8%*	*8.7%*	*13.6%*	*12.5%*	
% YoY	*17.5%*	*21.0%*	*35.1%*	*34.9%*	*27.6%*	*28.2%*	*25.7%*	*16.8%*	*17.7%*	*21.5%*	*27.1%*	*34.8%*	*38.1%*	*42.9%*	*36.2%*
DDR Demand (M 1Gb)					**324,263**					**394,034**					**536,665**
% YoY					*27.6%*					*21.5%*					*36.2%*
Inventory adjustment	-5,500	3,800	400	-900	-2,200	16,100	12,000	3,500	-2,400	29,200	-5,000	-10,000	-19,200	-33,000	-67,200
Inventory Adj. Demand (M 1Gb)	**64,379**	**76,866**	**87,629**	**93,190**	**322,064**	**105,674**	**103,832**	**105,363**	**108,367**	**423,234**	**108,879**	**113,806**	**121,479**	**125,302**	**469,466**
% QoQ	*-5.1%*	*19.4%*	*14.0%*	*6.3%*		*13.4%*	*-1.7%*	*1.5%*	*2.9%*		*0.5%*	*4.5%*	*6.7%*	*3.1%*	
% YoY	*4.0%*	*12.2%*	*29.5%*	*37.3%*	*21.1%*	*64.1%*	*35.1%*	*20.2%*	*16.3%*	*31.4%*	*3.0%*	*9.6%*	*15.3%*	*15.6%*	*10.9%*
Wafer out (k wpm, 12" equi.)	**1,722**	**1,772**	**1,816**	**1,857**	**1,792**	**1,887**	**1,907**	**1,935**	**1,966**	**1,924**	**2,010**	**2,063**	**2,115**	**2,168**	**2,089**
% QoQ	*2.4%*	*2.9%*	*2.5%*	*2.2%*		*1.6%*	*1.0%*	*1.5%*	*1.6%*		*2.3%*	*2.6%*	*2.5%*	*2.5%*	
% YoY	*22.4%*	*17.8%*	*13.7%*	*10.4%*	*15.8%*	*9.6%*	*7.6%*	*6.6%*	*5.9%*	*7.4%*	*6.5%*	*8.2%*	*9.3%*	*10.3%*	*8.6%*
% Sufficiency Ratio	***0.9%***	***-0.5%***	***-2.1%***	***-6.4%***	***-2.3%***	***-16.0%***	***-10.0%***	***-4.4%***	***-2.4%***	***-8.1%***	***-2.0%***	***-1.2%***	***-1.1%***	***-0.6%***	***-1.2%***
Blended ASP incl. HBM ($/Gb)	**0.41**	**0.42**	**0.46**	**0.61**	**0.48**	**1.09**	**1.65**	**2.01**	**2.25**	**1.78**	**2.49**	**2.65**	**2.77**	**2.83**	**2.69**
% QoQ	*-4.8%*	*1.9%*	*9.6%*	*32.2%*		*80.3%*	*50.6%*	*22.1%*	*12.1%*		*10.4%*	*6.3%*	*4.6%*	*2.2%*	
% YoY	*41.5%*	*21.8%*	*19.7%*	*40.4%*	*32.0%*	*166.0%*	*293.1%*	*338.3%*	*271.8%*	*270.7%*	*127.8%*	*60.8%*	*37.8%*	*25.6%*	*51.3%*
DDR ASP ($/Gb)	**0.33**	**0.35**	**0.38**	**0.53**	**0.40**	**1.05**	**1.65**	**2.03**	**2.30**	**1.78**	**2.48**	**2.63**	**2.74**	**2.79**	**2.67**
% QoQ	*-6.2%*	*7.0%*	*7.0%*	*42.0%*		*97.0%*	*57.0%*	*23.0%*	*13.0%*		*8.0%*	*6.0%*	*4.0%*	*2.0%*	
% YoY	*20.3%*	*10.6%*	*11.8%*	*52.5%*	*26.3%*	*220.3%*	*369.9%*	*440.2%*	*329.9%*	*341.4%*	*135.7%*	*59.1%*	*34.5%*	*21.4%*	*49.4%*
HBM ASP ($/Gb)	**1.42**	**1.55**	**1.56**	**1.52**	**1.52**	**1.55**	**1.56**	**1.79**	**1.89**	**1.72**	**2.54**	**2.76**	**2.98**	**3.08**	**2.86**
% QoQ	*2.7%*	*9.4%*	*0.4%*	*-2.5%*		*2.0%*	*0.4%*	*14.7%*	*5.7%*		*34.5%*	*8.7%*	*8.2%*	*3.1%*	
% YoY	*25.6%*	*28.3%*	*16.9%*	*10.0%*	*15.6%*	*9.2%*	*0.2%*	*14.5%*	*24.1%*	*13.4%*	*63.7%*	*77.2%*	*67.1%*	*63.0%*	*66.4%*

Source: Company data, UBS estimates

Figure 34: DRAM vendors' market share in bit shipments

	Q125	Q225	Q325	Q425	2025	Q126	Q226E	Q326E	Q426E	2026E	Q127E	Q227E	Q327E	Q427E	2027E
Samsung	39.3%	37.3%	38.2%	38.3%	38.2%	38.0%	37.9%	37.6%	37.3%	37.7%	37.7%	37.4%	37.9%	37.0%	37.5%
SK Hynix	28.9%	30.6%	29.5%	29.3%	29.6%	28.7%	29.3%	30.8%	31.2%	30.1%	29.5%	29.7%	29.4%	29.9%	29.6%
Micron	22.8%	23.3%	23.8%	23.6%	23.4%	24.3%	23.8%	22.9%	22.7%	23.4%	23.4%	23.3%	22.9%	22.9%	23.1%
Nanya Tech	1.0%	1.4%	1.6%	1.8%	1.5%	1.6%	1.5%	1.3%	1.1%	1.4%	1.2%	1.2%	1.3%	1.4%	1.3%
CXMT	7.7%	7.1%	6.6%	6.9%	7.0%	7.1%	7.3%	7.2%	7.5%	7.3%	8.1%	8.4%	8.4%	8.6%	8.4%
Others	0.3%	0.3%	0.2%	0.2%	0.3%	0.2%	0.2%	0.2%	0.2%	0.2%	0.2%	0.2%	0.2%	0.2%	0.2%
Total Supply	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**	**100.0%**

Source: Gartner, Company data, UBS estimates

Figure 35: DRAM vendors' bit shipments QoQ and YoY growth

	Q125 QoQ	Q225 QoQ	Q325 QoQ	Q425 QoQ	2025 YoY	Q126 QoQ	Q226E QoQ	Q326E QoQ	Q426E QoQ	2026E YoY	Q127E QoQ	Q226E QoQ	Q327E QoQ	Q427E QoQ	2027E YoY
Samsung	1.5%	11.5%	15.0%	2.0%	9.7%	1.0%	5.0%	6.8%	4.2%	21.8%	1.9%	4.5%	8.2%	1.3%	18.6%
SK Hynix	-7.0%	25.0%	8.0%	0.8%	24.3%	-0.3%	7.5%	13.4%	6.5%	25.7%	-4.7%	5.8%	5.9%	5.5%	17.5%
Micron	-8.0%	20.3%	15.0%	0.5%	21.8%	5.1%	3.2%	3.4%	4.0%	23.5%	4.0%	5.0%	5.0%	4.0%	17.9%
Nanya Tech	8.4%	68.0%	26.4%	11.9%	54.3%	-6.2%	-3.0%	-6.7%	-6.9%	15.3%	2.5%	5.6%	17.6%	14.9%	9.7%
CXMT	24.8%	8.3%	4.8%	5.1%	100.6%	6.0%	6.8%	7.6%	8.3%	27.8%	8.9%	9.4%	7.3%	5.8%	36.9%
Others	-4.8%	2.5%	2.4%	2.4%	3.1%	-2.3%	0.0%	2.4%	0.0%	2.4%	-2.3%	0.0%	2.4%	0.0%	0.0%
Bit supply Growth	**-1.9%**	**17.7%**	**12.3%**	**1.7%**	**21.1%**	**1.8%**	**5.3%**	**7.7%**	**5.0%**	**23.6%**	**0.8%**	**5.4%**	**6.8%**	**3.7%**	**19.3%**

Source: Gartner, Company data, UBS estimates

Figure 36: DRAM vendors' wafers out (k wpm, 12" equivalent)

	Q125	Q225	Q325	Q425	2025	Q126	Q226E	Q326E	Q426E	2026E	Q127E	Q227E	Q327E	Q427E	2027E
Samsung	663	667	668	674	668	676	678	685	689	682	695	707	719	731	713
SK Hynix	475	499	519	544	509	564	572	583	600	580	618	636	649	662	641
Micron	315	313	311	309	312	307	307	307	307	307	317	327	337	347	332
Nanya Tech	49	53	59	60	57	60	60	60	60	60	60	63	70	78	70
CXMT	170	190	210	220	198	230	240	250	260	245	270	280	290	300	285
Other	50	50	50	50	50	50	50	50	50	50	50	50	50	50	50
Total	**1,722**	**1,772**	**1,816**	**1,857**	**1,792**	**1,887**	**1,907**	**1,935**	**1,966**	**1,924**	**2,010**	**2,063**	**2,115**	**2,168**	**2,089**
% YoY	*22.4%*	*17.8%*	*13.7%*	*10.4%*	*15.8%*	*9.6%*	*7.6%*	*6.6%*	*5.9%*	*7.4%*	*6.5%*	*8.2%*	*9.3%*	*10.3%*	*8.6%*

Source: Gartner, Company data, UBS estimates

Figure 37: DRAM bit demand growth by application

	Q125 QoQ	Q225 QoQ	Q325 QoQ	Q425 QoQ	2025 YoY	Q126 QoQ	Q226E QoQ	Q326E QoQ	Q426E QoQ	2026E YoY	Q127E QoQ	Q226E QoQ	Q327E QoQ	Q427E QoQ	2027E YoY
Total PC DRAM	**-6.1%**	**13.8%**	**18.5%**	**3.0%**	**20.2%**	**-15.2%**	**-2.0%**	**1.2%**	**-1.4%**	**-4.6%**	**-10.3%**	**8.8%**	**17.1%**	**7.1%**	**5.0%**
Desktops	0.4%	10.8%	9.3%	5.6%	22.8%	-12.9%	-5.4%	-6.1%	0.2%	-9.7%	-3.9%	8.4%	15.2%	10.7%	8.3%
Notebooks	-8.1%	14.6%	20.7%	2.2%	24.3%	-15.3%	-0.2%	2.5%	-1.1%	-2.3%	-10.3%	9.6%	18.3%	7.7%	7.9%
Upgrade Modules	-8.1%	17.3%	27.5%	1.9%	-10.9%	-20.0%	-8.3%	12.1%	-8.3%	-8.7%	-30.0%	0.0%	10.0%	-18.6%	-32.2%
Servers	**5.4%**	**7.2%**	**35.8%**	**11.0%**	**57.2%**	**0.9%**	**8.2%**	**19.8%**	**9.5%**	**52.1%**	**16.5%**	**15.0%**	**17.7%**	**13.6%**	**73.8%**
Graphics	13.2%	13.4%	17.5%	-1.0%	26.8%	-9.7%	5.3%	13.4%	0.0%	10.5%	-8.4%	4.9%	13.1%	-0.2%	8.8%
Game Consoles	22.7%	4.2%	4.0%	1.9%	18.3%	0.3%	2.3%	4.2%	-8.5%	6.5%	2.3%	2.7%	4.3%	-3.4%	1.5%
HDTVs	-17.5%	1.8%	21.2%	5.8%	9.7%	-21.5%	1.3%	16.2%	14.5%	2.0%	-21.1%	3.4%	15.7%	12.2%	7.2%
Total Mobile DRAM	**-5.6%**	**-1.0%**	**14.1%**	**10.2%**	**15.6%**	**-13.2%**	**-6.2%**	**4.1%**	**12.7%**	**-0.6%**	**-7.5%**	**-2.9%**	**8.1%**	**16.0%**	**7.5%**
Smartphones	**-4.8%**	**-1.4%**	**14.6%**	**10.1%**	**16.7%**	**-13.2%**	**-7.0%**	**4.1%**	**12.9%**	**-1.2%**	**-6.8%**	**-3.5%**	**8.0%**	**16.6%**	**7.8%**
Apple	-31.8%	-5.6%	29.4%	80.5%	31.3%	-32.6%	-5.8%	4.7%	74.7%	28.6%	-32.1%	-5.8%	1.1%	87.0%	17.5%
Samsung	26.2%	-7.0%	11.5%	-2.8%	17.6%	8.3%	-1.4%	7.7%	-10.4%	9.9%	17.3%	-10.4%	13.3%	-11.1%	-17.8%
Non-Apple/Samsung	-3.0%	1.7%	11.8%	-6.1%	12.0%	-9.7%	-9.7%	2.2%	-3.7%	-15.2%	3.1%	1.2%	9.2%	-3.0%	14.0%
Tablet PCs	-16.0%	5.0%	8.9%	10.2%	3.2%	-13.1%	4.0%	5.0%	10.4%	6.7%	-16.2%	5.3%	8.4%	8.9%	3.3%
Others	4.3%	4.0%	2.8%	2.9%	12.2%	3.0%	2.6%	1.8%	7.3%	12.8%	-11.2%	2.7%	3.6%	8.8%	0.8%
Total Bit Demand	**0.2%**	**4.6%**	**19.4%**	**7.9%**	**27.6%**	**-4.8%**	**2.5%**	**10.9%**	**8.7%**	**21.5%**	**2.8%**	**8.7%**	**13.6%**	**12.5%**	**36.2%**

Source: Gartner, Company data, UBS estimates

Figure 38: DRAM content assumptions (MB/unit) by application

		Q125	Q225	Q325	Q425	2025	Q126	Q226E	Q326E	Q426E	2026E	Q127E	Q227E	Q327E	Q427E	2027E
Total PC		**12,038**	**12,569**	**12,982**	**13,302**	**12,766**	**13,312**	**13,548**	**13,745**	**13,965**	**13,637**	**14,008**	**14,214**	**14,697**	**15,231**	**14,579**
	% YoY	*9%*	*12%*	*15%*	*16%*	*13%*	*11%*	*8%*	*6%*	*5%*	*7%*	*5%*	*5%*	*7%*	*9%*	*7%*
Desktops		11,860	12,370	12,422	12,422	12,283	12,422	12,622	12,822	13,022	12,708	13,022	13,222	13,722	14,222	13,589
Notebooks		12,401	13,001	13,601	14,081	13,328	14,081	14,281	14,481	14,681	14,379	14,681	14,881	15,381	15,881	15,252
Upgrade Modules		9,900	10,100	10,300	10,500	10,232	10,500	10,700	10,900	11,100	10,794	11,100	11,100	11,100	11,300	11,144
Servers		**667,847**	**707,847**	**770,462**	**835,999**	**752,630**	**870,999**	**925,999**	**985,999**	**1,050,874**	**963,025**	**1,200,874**	**1,350,874**	**1,500,874**	**1,659,431**	**1,435,222**
	% YoY	*17%*	*20%*	*26%*	*34%*	*25%*	*30%*	*31%*	*28%*	*26%*	*28%*	*38%*	*46%*	*52%*	*58%*	*49%*
Smartphones		**7,739**	**7,829**	**8,255**	**8,611**	**8,126**	**8,664**	**8,707**	**8,977**	**9,327**	**8,923**	**9,239**	**9,105**	**9,313**	**10,007**	**9,434**
	% YoY	*11%*	*11%*	*13%*	*15%*	*13%*	*12%*	*11%*	*9%*	*8%*	*10%*	*7%*	*5%*	*4%*	*7%*	*6%*
Apple		7,600	7,700	8,940	9,590	8,652	9,690	9,690	10,400	10,800	10,227	11,000	11,000	11,400	12,800	11,711
Samsung		8,619	8,359	8,866	8,764	8,655	9,573	9,357	9,674	9,624	9,557	10,085	9,653	10,161	9,972	7,699
Non-Apple/Samsung		7,500	7,700	7,900	8,100	7,801	8,000	8,100	8,200	8,300	8,144	8,200	8,200	8,300	8,400	8,277

Source: Gartner, Company data, UBS estimates

Figure 39: UBS server DRAM bottom-up model

	2020	2021	2022	2023	2024	2025	2026E	2027E
Total Server shipments (K unit)	**13,550**	**13,967**	**14,943**	**14,087**	**16,025**	**20,078**	**23,874**	**27,844**
% YoY	***7.5%***	***3.1%***	***7.0%***	***-5.7%***	***13.8%***	***25.3%***	***18.9%***	***16.6%***
Traditional enterprise	**5,403**	**5,147**	**5,028**	**4,217**	**4,283**	**4,214**	**4,298**	**4,384**
% YoY	*-5.9%*	*-4.7%*	*-2.3%*	*-16.1%*	*1.6%*	*-1.6%*	*2.0%*	*2.0%*
% of total	*39.9%*	*36.8%*	*33.6%*	*29.9%*	*26.7%*	*21.0%*	*18.0%*	*15.7%*
Hyperscalers	**8,147**	**8,822**	**9,915**	**9,883**	**11,102**	**15,864**	**19,576**	**23,460**
% YoY	*25.2%*	*8.3%*	*12.4%*	*-0.3%*	*12.3%*	*42.9%*	*23.4%*	*19.8%*
% of total	*60.1%*	*63.2%*	*66.4%*	*70.2%*	*69.3%*	*79.0%*	*82.0%*	*84.3%*
Standard servers	**7,355**	**7,914**	**8,916**	**8,841**	**9,265**	**14,363**	**18,138**	**21,501**
% YoY	*23.8%*	*7.6%*	*12.7%*	*-0.8%*	*4.8%*	*55.0%*	*26.3%*	*18.5%*
% of total hyperscalers	*90.3%*	*89.7%*	*89.9%*	*89.5%*	*83.5%*	*90.5%*	*92.7%*	*91.6%*
% of total servers	*54.3%*	*56.7%*	*59.7%*	*62.8%*	*57.8%*	*71.5%*	*76.0%*	*77.2%*
AI Servers	**520**	**630**	**730**	**783**	**1,591**	**1,243**	**1,172**	**1,685**
% YoY	*73.3%*	*21.2%*	*15.9%*	*7.3%*	*103.1%*	*-21.8%*	*-5.7%*	*43.8%*
% of total hyperscalers	*6.4%*	*7.1%*	*7.4%*	*7.9%*	*14.3%*	*7.8%*	*6.0%*	*7.2%*
% of total servers	*3.8%*	*4.5%*	*4.9%*	*5.6%*	*9.9%*	*6.2%*	*4.9%*	*6.1%*
Database servers	**272**	**278**	**270**	**259**	**246**	**258**	**266**	**274**
% YoY	*2.0%*	*2.0%*	*-3.0%*	*-4.0%*	*-5.0%*	*5.0%*	*3.0%*	*3.0%*
% of total hyperscalers	*3.3%*	*3.1%*	*2.7%*	*2.6%*	*2.2%*	*1.6%*	*1.4%*	*1.2%*
% of total servers	*2.0%*	*2.0%*	*1.8%*	*1.8%*	*1.5%*	*1.3%*	*1.1%*	*1.0%*
Avg. DRAM content (GB)	**392**	**457**	**461**	**524**	**576**	**735**	**940**	**1,402**
% YoY	***32.5%***	***16.7%***	***0.9%***	***13.5%***	***9.9%***	***27.7%***	***27.9%***	***49.0%***
Traditional enterprise	**356**	**417**	**420**	**464**	**470**	**480**	**500**	**520**
% YoY	*27.4%*	*17.3%*	*0.7%*	*10.3%*	*1.4%*	*2.1%*	*4.2%*	*4.0%*
Hyperscalers Blended	**416**	**480**	**482**	**549**	**650**	**803**	**1,037**	**1,566**
% YoY	*27.3%*	*15.5%*	*0.4%*	*13.8%*	*18.4%*	*23.5%*	*29.2%*	*51.0%*
Standard servers	**385**	**430**	**432**	**459**	**464**	**480**	**515**	**555**
% YoY	*28.3%*	*11.7%*	*0.5%*	*6.3%*	*1.1%*	*3.4%*	*7.3%*	*7.8%*
AI servers	**600**	**840**	**860**	**1,348**	**1,647**	**4,435**	**9,048**	**14,493**
% YoY	*7.1%*	*40.0%*	*2.4%*	*56.8%*	*22.1%*	*169.3%*	*104.0%*	*60.2%*
Database Servers	**900**	**1,100**	**1,120**	**1,200**	**1,210**	**1,280**	**1,340**	**1,420**
% YoY	*36.4%*	*22.2%*	*1.8%*	*7.1%*	*0.8%*	*5.8%*	*4.7%*	*6.0%*
Total DRAM Demand (mn Gb)	**42,491**	**51,095**	**55,158**	**59,039**	**73,831**	**118,088**	**179,621**	**312,207**
% YoY	***42.3%***	***20.2%***	***8.0%***	***7.0%***	***25.1%***	***59.9%***	***52.1%***	***73.8%***
Total DDR Demand (mn Gb)	**42,491**	**51,095**	**55,158**	**56,355**	**65,224**	**100,602**	**146,482**	**253,560**
% YoY	***42.3%***	***20.2%***	***8.0%***	***2.2%***	***15.7%***	***54.2%***	***45.6%***	***73.1%***
Traditional enterprise	**15,382**	**17,191**	**16,908**	**15,641**	**16,106**	**16,181**	**17,193**	**18,238**
% YoY	*19.8%*	*11.8%*	*-1.6%*	*-7.5%*	*3.0%*	*0.5%*	*6.3%*	*6.1%*
Hyperscalers	**27,110**	**33,904**	**38,250**	**43,398**	**57,725**	**101,907**	**162,428**	**293,969**
% YoY	*59.3%*	*25.1%*	*12.8%*	*13.5%*	*33.0%*	*76.5%*	*59.4%*	*81.0%*
Standard servers	**22,652**	**27,225**	**30,813**	**32,465**	**34,393**	**55,152**	**74,728**	**95,464**
% YoY	*58.9%*	*20.2%*	*13.2%*	*5.4%*	*5.9%*	*60.4%*	*35.5%*	*27.7%*
AI servers	**2,496**	**4,234**	**5,022**	**8,448**	**20,952**	**44,111**	**84,850**	**195,394**
% YoY	*85.7%*	*69.6%*	*18.6%*	*68.2%*	*148.0%*	*110.5%*	*92.4%*	*130.3%*
HBM DRAM Demand				***2,684***	***8,607***	***17,486***	***33,139***	***58,648***
% YoY					*220.7%*	*103.2%*	*89.5%*	*77.0%*
DDR DRAM Demand				***6,081***	***12,345***	***26,625***	***51,711***	***136,746***
% YoY					*103.0%*	*115.7%*	*94.2%*	*164.4%*
LPDDR / SOCAMM Demand					***468***	***7,613***	***15,179***	***33,798***
% YoY						*1527.9%*	*99.4%*	*122.7%*
Database Servers	**1,962**	**2,445**	**2,415**	**2,484**	**2,380**	**2,643**	**2,850**	**3,111**
% YoY	*39.1%*	*24.7%*	*-1.2%*	*2.9%*	*-4.2%*	*11.1%*	*7.8%*	*9.1%*

Source: Dell'Oro Group, Gartner, Company data, UBS estimates

Figure 40: UBS bottom-up server procurement model (k units)

(K Units)	2022	2023	2024	2025	1Q26	2Q26E	3Q26E	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E
Quarterly														
Alphabet	1,039	1,085	1,247	1,845	530	600	580	640	2,350	720	720	760	750	2,950
AWS	2,237	2,131	2,276	3,087	940	960	1,020	1,000	3,920	1,150	1,180	1,250	1,240	4,820
Microsoft	1,056	1,274	1,468	2,457	680	760	830	820	3,090	850	920	1,000	970	3,740
Meta	2,192	2,027	2,188	3,409	950	1,080	1,100	1,050	4,180	1,200	1,280	1,250	1,260	4,990
Baidu	71	70	37	30	10	10	11	10	41	13	14	13	12	52
Alibaba	376	188	305	456	135	145	155	160	595	158	167	175	180	680
Tencent	118	149	303	268	75	82	88	82	327	82	92	100	95	369
ByteDance	358	550	770	1,306	360	420	430	440	1,650	450	480	520	490	1,940
Total Top 8	**7,448**	**7,473**	**8,594**	**12,858**	**3,680**	**4,057**	**4,214**	**4,202**	**16,153**	**4,623**	**4,853**	**5,068**	**4,997**	**19,541**
% YoY	*13.5%*	*0.3%*	*15.0%*	*49.6%*	*28.2%*	*31.7%*	*25.0%*	*18.8%*	*25.6%*	*25.6%*	*19.6%*	*20.3%*	*18.9%*	*21.0%*
% QoQ					*4.1%*	*10.2%*	*3.9%*	*-0.3%*		*10.0%*	*5.0%*	*4.4%*	*-1.4%*	
Others	2,468	2,410	2,901	1,879	787	839	889	908	3,423	922	999	1,021	977	3,919
Total hyperscale	**9,915**	**9,883**	**11,494**	**14,737**	**4,467**	**4,896**	**5,103**	**5,110**	**19,576**	**5,545**	**5,852**	**6,089**	**5,974**	**23,460**
% YoY	*12.4%*	*-0.3%*	*16.3%*	*28.2%*	*38.8%*	*51.2%*	*26.5%*	*20.3%*	*32.8%*	*24.1%*	*19.5%*	*19.3%*	*16.9%*	*19.8%*
% QoQ					*5.2%*	*9.6%*	*4.2%*	*0.1%*		*8.5%*	*5.5%*	*4.0%*	*-1.9%*	
YoY (%)														
Alphabet	*24.0%*	*4.4%*	*15.0%*	*48.0%*	*51.8%*	*6.8%*	*26.9%*	*34.1%*	*27.3%*	*35.8%*	*20.0%*	*31.0%*	*17.2%*	*25.5%*
AWS	*24.5%*	*-4.7%*	*6.8%*	*35.6%*	*40.0%*	*63.4%*	*20.0%*	*2.3%*	*27.0%*	*22.3%*	*22.9%*	*22.5%*	*24.0%*	*23.0%*
Microsoft	*-2.9%*	*20.6%*	*15.3%*	*67.4%*	*17.9%*	*29.3%*	*20.4%*	*35.9%*	*25.7%*	*25.0%*	*21.1%*	*20.5%*	*18.3%*	*21.0%*
Meta	*20.7%*	*-7.5%*	*8.0%*	*55.8%*	*23.0%*	*21.3%*	*23.9%*	*22.4%*	*22.6%*	*26.3%*	*18.5%*	*13.6%*	*20.0%*	*19.4%*
Baidu	*-34.0%*	*-1.6%*	*-47.9%*	*-16.5%*	*44.8%*	*42.5%*	*3.4%*	*68.6%*	*34.5%*	*30.0%*	*40.0%*	*18.2%*	*20.0%*	*26.8%*
Alibaba	*-10.6%*	*-50.2%*	*62.5%*	*49.6%*	*81.9%*	*10.3%*	*41.3%*	*13.8%*	*30.5%*	*17.0%*	*15.2%*	*12.9%*	*12.5%*	*14.3%*
Tencent	*-56.1%*	*25.9%*	*103.3%*	*-11.6%*	*2.3%*	*42.6%*	*103.2%*	*-12.2%*	*22.2%*	*9.3%*	*12.2%*	*13.6%*	*15.9%*	*12.8%*
ByteDance	*60.6%*	*53.8%*	*39.9%*	*69.7%*	*4.0%*	*63.3%*	*32.9%*	*16.0%*	*26.3%*	*25.0%*	*14.3%*	*20.9%*	*11.4%*	*17.6%*

Source: Dell'Oro Group, company data, UBS estimates

Figure 41: UBS NAND Flash supply/demand summary

	Q125	Q225	Q325	Q425	2025	Q126	Q226E	Q326E	Q426E	2026E	Q127E	Q227E	Q327E	Q427E	2027E
Revenues ($ mil)	**12,302**	**14,726**	**16,969**	**21,320**	**65,317**	**37,952**	**66,964**	**90,474**	**104,858**	**300,248**	**114,673**	**123,798**	**135,805**	**140,845**	**515,120**
% QoQ	*-20.1%*	*19.7%*	*15.2%*	*25.6%*		*78.0%*	*76.4%*	*35.1%*	*15.9%*		*9.4%*	*8.0%*	*9.7%*	*3.7%*	
% YoY	*-1.0%*	*-1.1%*	*4.9%*	*38.5%*	*10.9%*	*208.5%*	*354.7%*	*433.2%*	*391.8%*	*359.7%*	*202.2%*	*84.9%*	*50.1%*	*34.3%*	*71.6%*
Supply (Shipments Based) (M 1GB)	**201,828**	**247,795**	**271,936**	**273,324**	**994,882**	**270,305**	**285,595**	**301,455**	**314,758**	**1,172,113**	**330,982**	**340,303**	**362,436**	**375,886**	**1,409,606**
% QoQ	*-6.0%*	*22.8%*	*9.7%*	*0.5%*		*-1.1%*	*5.7%*	*5.6%*	*4.4%*		*5.2%*	*2.8%*	*6.5%*	*3.7%*	
% YoY	*-6.3%*	*15.1%*	*25.6%*	*27.3%*	*15.4%*	*33.9%*	*15.3%*	*10.9%*	*15.2%*	*17.8%*	*22.4%*	*19.2%*	*20.2%*	*19.4%*	*20.3%*
Demand (End Consumption) (M 1GB)	**235,552**	**242,791**	**280,969**	**321,309**	**1,080,621**	**298,176**	**303,627**	**327,635**	**367,018**	**1,296,456**	**354,680**	**371,823**	**408,221**	**460,507**	**1,595,232**
% QoQ	*-8.1%*	*3.1%*	*15.7%*	*14.4%*		*-7.2%*	*1.8%*	*7.9%*	*12.0%*		*-3.4%*	*4.8%*	*9.8%*	*12.8%*	
% YoY	*11.4%*	*13.0%*	*20.1%*	*25.4%*	*17.9%*	*26.6%*	*25.1%*	*16.6%*	*14.2%*	*20.0%*	*18.9%*	*22.5%*	*24.6%*	*25.5%*	*23.0%*
Inventory adjustment	-39,600	3,700	-6,200	-33,500	-75,600	23,500	26,500	-8,000	-45,500	-3,500	-20,500	-28,000	-43,200	-84,500	-176,200
Inventory Adj. Demand (M 1GB)	**195,952**	**246,491**	**274,769**	**287,809**	**1,005,021**	**321,676**	**330,127**	**319,635**	**321,518**	**1,292,956**	**334,180**	**343,823**	**365,021**	**376,007**	**1,419,032**
% QoQ	*-8.0%*	*25.8%*	*11.5%*	*4.7%*		*11.8%*	*2.6%*	*-3.2%*	*0.6%*		*3.9%*	*2.9%*	*6.2%*	*3.0%*	
% YoY	*-14.3%*	*9.9%*	*24.6%*	*35.1%*	*13.4%*	*64.2%*	*33.9%*	*16.3%*	*11.7%*	*28.6%*	*3.9%*	*4.1%*	*14.2%*	*16.9%*	*9.8%*
Wafer out (k wpm, 12" equi.)	**1,418**	**1,288**	**1,268**	**1,313**	**1,322**	**1,366**	**1,343**	**1,338**	**1,351**	**1,349**	**1,361**	**1,357**	**1,353**	**1,349**	**1,355**
% QoQ	*-0.4%*	*-9.2%*	*-1.5%*	*3.5%*		*4.1%*	*-1.7%*	*-0.4%*	*1.0%*		*0.7%*	*-0.3%*	*-0.3%*	*-0.3%*	
% YoY	*23.7%*	*2.5%*	*-5.8%*	*-7.8%*	*2.2%*	*-3.6%*	*4.3%*	*5.5%*	*2.9%*	*2.1%*	*-0.4%*	*1.0%*	*1.1%*	*-0.2%*	*0.4%*
% Sufficiency Ratio	***3.0%***	***0.5%***	***-1.0%***	***-5.0%***	***-1.0%***	***-16.0%***	***-13.5%***	***-5.7%***	***-2.1%***	***-9.3%***	***-1.0%***	***-1.0%***	***-0.7%***	***0.0%***	***-0.7%***
NAND ASP ($/1GB)	**0.06**	**0.06**	**0.06**	**0.08**	**0.07**	**0.14**	**0.23**	**0.30**	**0.33**	**0.26**	**0.35**	**0.36**	**0.37**	**0.37**	**0.37**
% QoQ	*-15.0%*	*-2.5%*	*5.0%*	*25.0%*		*80.0%*	*67.0%*	*28.0%*	*11.0%*		*4.0%*	*5.0%*	*3.0%*	*0.0%*	
% YoY	*5.8%*	*-14.1%*	*-16.5%*	*8.8%*	*-3.9%*	*130.3%*	*294.5%*	*381.0%*	*327.1%*	*290.2%*	*146.8%*	*55.2%*	*24.8%*	*12.5%*	*42.7%*

Source: Company data, UBS estimates

Figure 42: NAND Flash vendors' market share in bit shipments

	Q125	Q225	Q325	Q425	2025	Q126	Q226E	Q326E	Q426E	2026E	Q127E	Q227E	Q327E	Q427E	2027E
Samsung	31.0%	31.8%	31.8%	28.5%	30.7%	31.0%	29.9%	29.0%	29.0%	29.7%	29.1%	29.1%	28.8%	28.2%	28.8%
Kioxia	15.3%	12.5%	15.5%	16.2%	14.9%	14.7%	14.6%	14.9%	15.0%	14.8%	16.0%	14.8%	14.5%	14.7%	15.0%
SanDisk	13.2%	11.2%	11.7%	12.0%	12.0%	11.6%	11.5%	12.5%	12.0%	11.9%	10.8%	11.1%	11.9%	12.3%	11.6%
SK Hynix	12.8%	18.2%	15.7%	17.1%	16.1%	15.5%	16.9%	16.5%	16.7%	16.4%	16.2%	16.8%	16.7%	16.5%	16.6%
Micron	14.1%	14.3%	12.6%	13.1%	13.5%	13.7%	13.6%	13.7%	13.9%	13.8%	13.6%	13.7%	13.2%	13.2%	13.4%
YMTC	12.5%	11.3%	11.9%	12.2%	12.0%	12.7%	12.6%	12.6%	12.7%	12.7%	13.5%	14.0%	14.1%	14.4%	14.0%
Total Bit Supply	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**	**100%**

Source: Gartner, company data, UBS estimates

Figure 43: NAND Flash bit shipments QoQ and YoY growth

	Q125 QoQ	Q225 QoQ	Q325 QoQ	Q425 QoQ	2025 YoY	Q126 QoQ	Q226E QoQ	Q327E QoQ	Q428E QoQ	2026E YoY	Q127E QoQ	Q227E QoQ	Q327E QoQ	Q427E QoQ	2027E YoY
Samsung	-10.0%	25.9%	10.0%	-10.0%	3.0%	7.5%	2.0%	2.3%	4.2%	13.8%	5.8%	2.6%	5.6%	1.7%	16.7%
Kioxia	-10.0%	0.0%	36.0%	5.0%	13.5%	-10.0%	5.0%	8.0%	5.0%	17.4%	11.7%	-5.0%	5.0%	5.0%	21.4%
SanDisk	-6.0%	4.0%	15.0%	3.0%	17.7%	-5.0%	5.0%	15.0%	0.0%	17.0%	-5.0%	5.0%	15.0%	7.0%	16.9%
SK Hynix	-16.9%	73.6%	-5.0%	9.6%	11.8%	-10.4%	15.0%	2.7%	5.8%	19.8%	2.3%	6.5%	6.0%	2.5%	21.5%
Micron	3.5%	24.3%	-3.3%	5.0%	25.6%	3.4%	5.0%	6.0%	6.0%	20.4%	3.0%	3.0%	3.0%	3.4%	17.2%
YMTC	15.7%	10.6%	15.5%	3.5%	57.3%	2.3%	5.5%	5.3%	5.2%	24.5%	12.0%	6.2%	7.6%	5.5%	33.1%
Total Bit Supply	**-6.0%**	**22.8%**	**9.7%**	**0.5%**	**15.4%**	**-1.1%**	**5.7%**	**5.6%**	**4.4%**	**17.8%**	**5.2%**	**2.8%**	**6.5%**	**3.7%**	**20.3%**

Source: Gartner, company data, UBS estimates

Figure 44: NAND Flash vendors wafers out (k wpm, 12" equivalent)

	Q125	Q225	Q325	Q425	2025	Q126	Q226E	Q326E	Q426E	2026E	Q127E	Q227E	Q327E	Q427E	2027E
Samsung	471	449	423	409	438	412	405	400	400	404	398	393	388	383	390
Kioxia / SanDisk	470	400	400	441	428	480	470	460	450	465	450	450	450	450	450
SK Hynix	216	189	185	186	194	185	178	183	201	186	208	204	200	196	202
Micron	132	111	111	117	118	130	130	130	130	130	130	130	130	130	130
YMTC	130	140	150	160	145	160	160	165	170	164	175	180	185	190	183
Total	**1,418**	**1,288**	**1,268**	**1,313**	**1,322**	**1,366**	**1,343**	**1,338**	**1,351**	**1,349**	**1,361**	**1,357**	**1,353**	**1,349**	**1,355**
% YoY	*23.7%*	*2.5%*	*-5.8%*	*-7.8%*	*2.2%*	*-3.6%*	*4.3%*	*5.5%*	*2.9%*	*2.1%*	*-0.4%*	*1.0%*	*1.1%*	*-0.2%*	*0.4%*

Source: Gartner, company data, UBS estimates

Figure 45: NAND Flash bit demand growth by application

	Q125 QoQ	Q225 QoQ	Q325 QoQ	Q425 QoQ	2025 YoY	Q126 QoQ	Q226E QoQ	Q327E QoQ	Q428E QoQ	2026E YoY	Q127E QoQ	Q227E QoQ	Q327E QoQ	Q427E QoQ	2027E YoY
Total Embedded	**-8.3%**	**3.1%**	**16.1%**	**14.8%**	**18.5%**	**-7.2%**	**1.9%**	**8.1%**	**12.3%**	**20.7%**	**-3.3%**	**5.0%**	**10.0%**	**13.1%**	**23.9%**
Smartphones	**-11.3%**	**-6.3%**	**17.3%**	**26.6%**	**16.9%**	**-21.7%**	**-6.2%**	**2.1%**	**30.7%**	**2.1%**	**-16.7%**	**-4.6%**	**6.2%**	**32.4%**	**8.3%**
iPhones	-29.0%	-7.7%	24.8%	69.7%	28.1%	-33.6%	-5.0%	-1.6%	70.4%	15.5%	-32.8%	-5.0%	1.3%	73.4%	9.5%
Samsung	42.2%	-13.4%	10.1%	-8.2%	15.0%	16.2%	-7.4%	7.5%	-12.7%	4.2%	32.6%	-15.9%	10.4%	-12.3%	8.9%
Non- iPhone/Samsung	-5.6%	-0.9%	13.2%	-5.0%	5.5%	-15.1%	-7.4%	4.7%	-1.4%	-16.6%	-2.8%	4.9%	11.7%	-0.8%	5.7%
Tablet PCs	-21.6%	10.0%	9.5%	17.2%	6.4%	-20.2%	12.0%	3.9%	17.5%	11.0%	-24.4%	13.1%	10.6%	14.3%	6.9%
SSD-Total	**-6.9%**	**7.5%**	**18.9%**	**11.2%**	**21.8%**	**-0.3%**	**4.0%**	**11.3%**	**6.9%**	**31.7%**	**3.8%**	**8.2%**	**12.1%**	**9.1%**	**33.7%**
SSD (notebooks)	-13.7%	9.3%	15.3%	-1.3%	9.0%	-14.1%	-0.3%	2.4%	0.0%	-6.2%	-10.2%	9.4%	15.7%	6.7%	6.9%
SSD (servers)	-4.0%	9.4%	36.0%	22.6%	45.1%	0.4%	5.5%	16.4%	8.3%	56.3%	9.2%	9.2%	12.8%	9.1%	47.0%
SSD (storage)	10.5%	7.9%	8.0%	10.1%	47.9%	15.2%	8.6%	8.2%	7.8%	47.7%	7.5%	9.7%	9.2%	8.8%	38.3%
SSD (others)	-14.6%	1.4%	-0.5%	0.2%	-12.8%	1.3%	-0.5%	8.8%	8.8%	8.0%	-6.0%	0.2%	9.4%	12.2%	12.5%
Total Flash cards	**-0.4%**	**1.6%**	**2.2%**	**-3.4%**	**-0.4%**	**-5.4%**	**-2.8%**	**-1.6%**	**-2.9%**	**-9.8%**	**-8.1%**	**-6.8%**	**-4.4%**	**-10.8%**	**-20.0%**
Mobile Phones - Flash cards	-6.3%	-1.5%	5.3%	-1.6%	-0.8%	-3.7%	-6.8%	1.5%	-7.4%	-8.8%	7.1%	-0.3%	6.7%	-6.1%	1.4%
Total Bit Demand	**-8.1%**	**3.1%**	**15.7%**	**14.4%**	**17.9%**	**-7.2%**	**1.8%**	**7.9%**	**12.0%**	**20.0%**	**-3.4%**	**4.8%**	**9.8%**	**12.8%**	**23.0%**

Source: Gartner, company data, UBS estimates

Figure 46: NAND Flash content assumptions (GB/unit) by application

	Q125	Q225	Q325	Q425	2025	Q126	Q226E	Q326E	Q426E	2026E	Q127E	Q227E	Q327E	Q427E	2027E
Smartphones - Blended	**198**	**193**	**216**	**251**	**216**	**223**	**224**	**227**	**276**	**238**	**243**	**235**	**238**	**293**	**253**
% YoY	*14%*	*14%*	*11%*	*22%*	*15%*	*12%*	*16%*	*5%*	*10%*	*10%*	*9%*	*5%*	*5%*	*6%*	*6%*
iPhones	517	512	573	579	538	576	581	586	594	572	599	604	627	653	610
Samsung	225	203	213	198	210	232	213	220	213	220	253	227	233	226	235
Non- iPhone/Samsung	127	127	132	137	131	127	132	137	142	134	132	137	142	146	139
SSD - Notebook PCs	**643**	**639**	**674**	**676**	**659**	**663**	**658**	**686**	**706**	**678**	**695**	**681**	**710**	**734**	**706**
% YoY	*2%*	*3%*	*2%*	*0%*	*2%*	*3%*	*3%*	*2%*	*4%*	*3%*	*5%*	*3%*	*4%*	*4%*	*4%*
SSD - Servers	**3,433**	**3,775**	**4,117**	**4,780**	**4,083**	**4,877**	**4,975**	**5,073**	**5,268**	**5,060**	**5,561**	**5,854**	**6,147**	**6,440**	**6,019**
% YoY	*12%*	*23%*	*34%*	*55%*	*33%*	*42%*	*32%*	*23%*	*10%*	*24%*	*14%*	*18%*	*21%*	*22%*	*19%*
SSD - Storage	**12,383**	**13,359**	**14,434**	**15,885**	**14,015**	**16,862**	**17,838**	**18,815**	**19,791**	**18,357**	**20,768**	**21,744**	**22,721**	**23,697**	**22,287**
% YoY	*57%*	*47%*	*41%*	*39%*	*45%*	*36%*	*34%*	*30%*	*25%*	*31%*	*23%*	*22%*	*21%*	*20%*	*21%*

Source: Gartner, company data, UBS estimates

Valuation Method and Risk Statement

Investing in the technology sector involves a high degree of risk. Rapid technological changes, increasing competition and exposure to macroeconomic cycles are among the many risks faced by investors in technology stocks. Moreover, it is extremely difficult to project the financial results of tech companies since their operating models are highly volatile and unpredictable. Finally, valuing technology stocks can prove challenging, as neither traditional nor non-traditional valuation measures have provided much insight into how these stocks trade.

Samsung: Some of Samsung's operating profit comes from mobile phones directly (and more indirectly for components) where we see revenue growth for the industry decelerating as the smartphone market starts to mature. The industry is characterized by product cycles, intense competition and requires significant scale and ongoing expenses for marketing/promotional activities. Some of its divisions (Memory and Display Panels) operate in industries which are cyclical and characterized by fluctuations in supply, demand and capital spending. Both industries have commodity-like features – although it is to be noted that the DRAM industry is now operating with significantly fewer players than before. Whilst overall cyclical risks have decreased longer term for DRAM (our "Goldilocks scenario"), cyclical corrections will likely occur. Consumer exposure overall at Samsung is universally high across products, although well balanced in terms of geographies. Strength in the Korean Won vs. major currencies can penalize earnings. Being a large component of the KOSPI, Samsung faces country risks as well indirectly. Corporate governance risks re the Samsung group center around some degree of opacity in decision-making, and limited consideration for shareholder returns (this may change). On the flipside, Samsung Electronics is the "crown jewel" of the group and its interests tend to supersede those of other Samsung companies. We value Samsung Electronics ordinary shares using a 12-month forward target P/BV multiple based on long-term average ROE and cost of equity. Our GDR price target comes from using a conversion ratio of 25 from Samsung ordinary shares (005930.KS) and the latest FX.

SK Hynix: Whilst overall cyclical risks have decreased longer term for DRAM (our "Goldilocks scenario"), amidst significant consolidation (3 players, from 15 in 1998) and lower growth, cyclical corrections will still occur as the sector retains commodity-like characteristics. The NAND flash industry structure has not changed as dramatically, but the coming rollout of 3D NAND flash manufacturing is restraining the players' willingness to add large increments of capacity at 2D. That being said, both segments are sensitive in particular to smartphones/tablets demand, mix within those segments, and to a lesser extent enterprise spend (enterprise solid state drives, servers). The industry also remains capital-intensive, and with contract prices negotiated every 2 weeks to 3 months, cash flows can change quickly – so adequate capitalization is required. Lastly, the industry may have to move to new memory technologies longer term to supplement, if not replace, current silicon-based Flash and RAM memory structures. We value SK Hynix shares using a 12-month forward target P/BV multiple based on long-term average ROE and cost of equity.

Micron: Memory industry is highly volatile. To arrive at our PT we use a SOTP valuation method. Memory has been fairly correlated to global GDP. Therefore from a macro perspective there could be potential downside to our PT should economic conditions worsen and leading to demand destruction. Should ASP declines be more severe than our expectations, there could be more downside in the stock. Scaling both DRAM and NAND is highly complex and risks on the Technology front abound. There could be further downside should technology migration be slower than we expect. On the upside, while we expect ASPs to decline on account of incremental supply addition, any delays to the timelines could moderate the growth of supply and create an upward bias to ASPs and MU. Other risks include a faster than expected adoption of AI driving up memory content in servers and cars.

Nanya Tech: Nanya Tech is a pure-play DRAM manufacturer based in Taiwan mainly focused on legacy DDR products for customers across various end applications. We think downside risks to our thesis could include a potential slowdown in end demand and hence the conventional memory cycle as well as Nanya's legacy-centric product portfolio. We value Nanya Tech shares using a 12-month forward target P/BV multiple based on long-term average ROE and cost of equity.

Kioxia: Our price target is based on PBR. Risk factors include: 1) strong cyclicality of the NAND market and potential price declines following supply-demand equilibrium, 2) reduced demand due to a slowdown in AI investment and restrained semiconductor investment, 3) uncertainty regarding the sustainability of technological superiority, and 4) high profit leverage during periods of price increases.

Required Disclosures

This document has been prepared by UBS Securities Pte. Ltd., an affiliate of UBS AG. UBS AG, its subsidiaries, branches and affiliates, including former Credit Suisse AG and its subsidiaries, branches and affiliates are referred to herein as "UBS".

For information on the ways in which UBS manages conflicts and maintains independence of its UBS Global Research product; historical performance information; certain additional disclosures concerning UBS Global Research recommendations; and terms and conditions for certain third party data used in research report, please visit https://www.ubs.com/disclosures. Unless otherwise indicated, information and data in this report are based on company disclosures including but not limited to annual, interim, quarterly reports and other company announcements. The figures contained in performance charts refer to the past; past performance is not a reliable indicator of future results. Additional information will be made available upon request. UBS Securities Co. Limited is licensed to conduct securities investment consultancy businesses by the China Securities Regulatory Commission. UBS acts or may act as principal in the debt securities (or in related derivatives) that may be the subject of this report. This recommendation was finalized on: 03 July 2026 10:21 AM GMT. UBS has designated certain UBS Global Research department members as Derivatives Research Analysts where those department members publish research principally on the analysis of the price or market for a derivative, and provide information reasonably sufficient upon which to base a decision to enter into a derivatives transaction. Where Derivatives Research Analysts co-author research reports with Equity Research Analysts or Economists, the Derivatives Research Analyst is responsible for the derivatives investment views, forecasts, and/or recommendations. **Quantitative Research Review:** UBS Global Research publishes a quantitative assessment of its analysts' responses to certain questions about the likelihood of an occurrence of a number of short term factors in a product known as the 'Quantitative Research Review'. Views contained in this assessment on a particular stock reflect only the views on those short term factors which are a different timeframe to the 12-month timeframe reflected in any equity rating set out in this note. For the latest responses, please see the Quantitative Research Review Addendum at the back of this report, where applicable. For previous responses please make reference to (i) previous UBS Global Research reports; and (ii) where no applicable research report was published that month, the Quantitative Research Review which can be found at https://neo.ubs.com/quantitative, or contact your UBS sales representative for access to the report or the Quantitative Research Team on ubs-quant-answers@ubs.com. A consolidated report which contains all responses is also available and again you should contact your UBS sales representative for details and pricing or the Quantitative Research team on the email above.

Analyst Certification:
Each research analyst primarily responsible for the content of this research report, in whole or in part, certifies that with respect to each security or issuer that the analyst covered in this report: (1) all of the views expressed accurately reflect his or her personal views about those securities or issuers and were prepared in an independent manner, including with respect to UBS, and (2) no part of his or her compensation was, is, or will be, directly or indirectly, related to the specific recommendations or views expressed by that research analyst in the research report.

UBS Global Research: Global Equity Rating Definitions

12-Month Rating	Definition	Coverage[1]	IB Services[2]
Buy	FSR is > 6% above the MRA.	54%	24%
Neutral	FSR is between -6% and 6% of the MRA.	40%	21%
Sell	FSR is > 6% below the MRA.	6%	21%

Source: UBS. Rating allocations are as of 31 March 2026.
1:Percentage of companies under coverage globally within the 12-month rating category.
2:Percentage of companies within the 12-month rating category for which investment banking (IB) services were provided within the past 12 months.

KEY DEFINITIONS:Forecast Stock Return (FSR) is defined as expected percentage price appreciation plus gross dividend yield over the next 12 months. In some cases, this yield may be based on accrued dividends. **Market Return Assumption (MRA)** is defined as the one-year local market interest rate plus 5% (a proxy for, and not a forecast of, the equity risk premium). **Under Review (UR)** Stocks may be flagged as UR by the analyst, indicating that the stock's price target and/or rating are subject to possible change in the near term, usually in response to an event that may affect the investment case or valuation. **Equity Price Targets** have an investment horizon of 12 months.

EXCEPTIONS AND SPECIAL CASES:UK and European Investment Fund ratings and definitions are: Buy: Positive on factors such as structure, management, performance record, discount; **Neutral:** Neutral on factors such as structure, management, performance record, discount; **Sell:** Negative on factors such as structure, management, performance record, discount. **Core Banding Exceptions (CBE):** Exceptions to the standard +/-6% bands may be granted by the Investment Review Consultation (IRC). Factors considered by the IRC include the stock's volatility and the credit spread of the respective company's debt. As a result, stocks deemed to be very high or low risk may be subject to higher or lower bands as they relate to the rating. When such exceptions apply, they will be identified in the Company Disclosures table in the relevant research piece.

Research analysts contributing to this report who are employed by any non-US affiliate of UBS Securities LLC are not registered/ qualified as research analysts with FINRA. Such analysts may not be associated persons of UBS Securities LLC and therefore are not subject to the FINRA restrictions on communications with a subject company, public appearances, and trading securities held by a research analyst account. The name of each affiliate and analyst employed by that affiliate contributing to this report, if any, follows.

UBS AG, Singapore Branch: Jimmy Yoon, Nicolas Gaudois.**UBS AG London Branch:** Francois-Xavier Bouvignies.**UBS Securities Co. Limited:** Jimmy Yu.**UBS Securities Japan Co., Ltd.:** Atsuhiro Kinoshita, Kenji Yasui.**UBS Securities LLC:** Gianmarco Vella, Timothy Arcuri.**UBS Securities Pte. Ltd., Taipei Branch:** Randy Abrams, Sunny Lin.

Company Disclosures

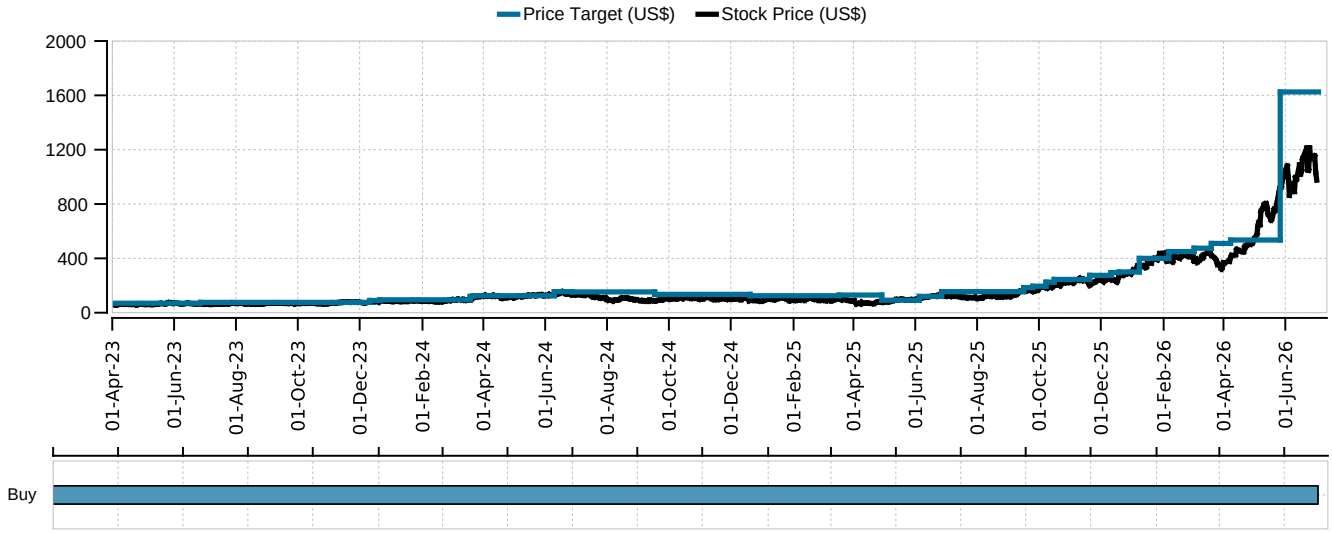
Company Name	Reuters	12-month rating	Price	Price date
Kioxia Holdings Corp[28b]	285A.T	Buy	¥83,300	03 Jul 2026
Micron Technology Inc[16,28b]	MU.O	Buy	US$975.56	02 Jul 2026
Nanya Technology	2408.TW	Buy	NT$409.50	03 Jul 2026
SK Hynix[8,28a]	000660.KS	Buy	Won2,425,000	03 Jul 2026
Samsung Electronics[2,6,8,28a]	005930.KS	Buy	Won309,500	03 Jul 2026

Source: UBS Global Research; LSEG Eikon. All prices as of local market close. Ratings in this table are the most current published ratings prior to this report. They may be more recent than the stock pricing date.

2. UBS has acted as manager/co-manager in the underwriting or placement of securities of this company/entity or one of its affiliates within the past 12 months.
6. This company/entity is, or within the past 12 months has been, a client of UBS Securities LLC and/or its affiliates, and investment banking services are being, or have been, provided.
8. The equity analyst covering this company, a member of his or her team, or one of their household members has a long common stock position in this company.
16. UBS Securities LLC makes a market in the securities and/or ADRs of this company.
28a. UBS holds a long position of 0.5% or more of the listed shares of this company.
28b. UBS holds a short position of 0.5% or more of the listed shares of this company.

Unless otherwise indicated, please refer to the Valuation and Risk sections within the body of this report. For a complete set of disclosure statements associated with the companies discussed in this report, including information on valuation and risk, please contact UBS Securities LLC, 11 Madison Avenue, New York, NY 10010, USA, Attention: Investment Research.

Micron Technology Inc (US$)

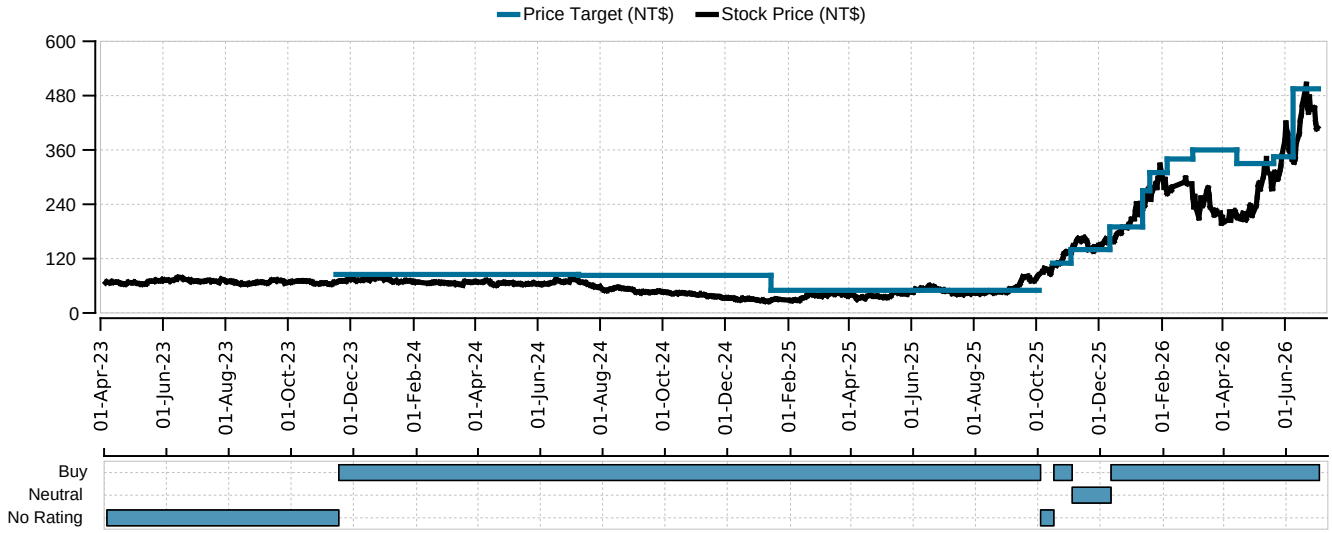


Date	Stock Price (US$)	Price Target (US$)	Rating
2023-03-31	60.34	70.00	Buy
2023-06-26	65.45	76.00	Buy
2023-12-10	74.96	90.00	Buy
2023-12-20	78.69	95.00	Buy
2024-03-18	93.78	120.00	Buy
2024-03-21	109.85	125.00	Buy

Date	Stock Price (US$)	Price Target (US$)	Rating
2024-06-09	130.94	155.00	Buy
2024-06-27	132.23	153.00	Buy
2024-09-16	87.18	135.00	Buy
2024-12-19	87.09	125.00	Buy
2025-03-17	103.11	130.00	Buy
2025-04-28	78.56	92.00	Buy
2025-06-04	103.25	120.00	Buy
2025-06-26	126.00	155.00	Buy
2025-09-15	157.77	185.00	Buy
2025-09-24	161.71	195.00	Buy
2025-10-07	185.69	225.00	Buy
2025-10-15	191.94	245.00	Buy
2025-11-19	225.92	275.00	Buy
2025-12-10	263.71	295.00	Buy
2025-12-18	248.55	300.00	Buy
2026-01-07	339.55	400.00	Buy
2026-02-05	382.89	450.00	Buy
2026-03-02	412.67	475.00	Buy
2026-03-19	444.27	510.00	Buy
2026-04-07	377.58	535.00	Buy
2026-05-26	895.88	1625.00	Buy

Source: UBS Global Research; LSEG Eikon as of 02-Jul-2026. All prices as of local market close. Ratings as of date shown.

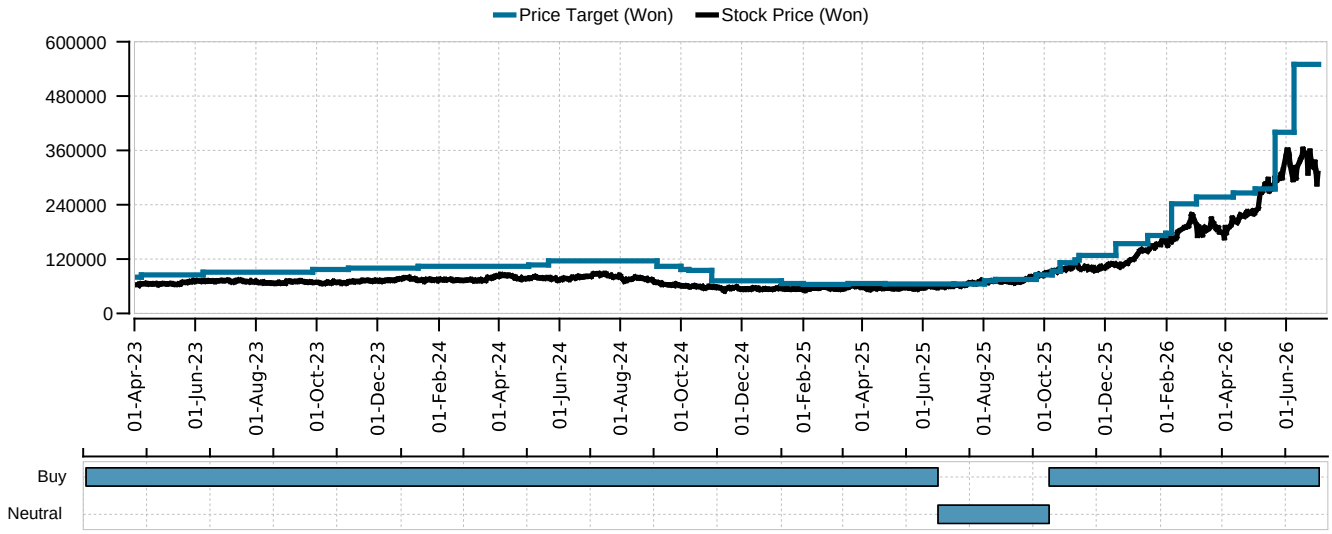
Nanya Technology (NT$)



Date	Stock Price (NT$)	Price Target (NT$)	Rating
2023-04-03	NaN	-	No Rating
2023-11-16	67.20	85.00	Buy
2024-07-10	72.30	83.00	Buy
2025-01-14	26.65	50.00	Buy
2025-10-03	83.00	-	No Rating
2025-10-16	94.90	110.00	Buy
2025-11-03	137.00	140.00	Neutral
2025-12-11	156.00	190.00	Buy
2026-01-12	239.00	270.00	Buy
2026-01-19	275.00	310.00	Buy
2026-02-05	275.00	340.00	Buy
2026-03-02	285.00	360.00	Buy
2026-04-14	220.00	330.00	Buy
2026-05-20	275.00	345.00	Buy
2026-06-08	339.50	495.00	Buy

Source: UBS Global Research; LSEG Eikon as of 03-Jul-2026. All prices as of local market close. Ratings as of date shown.

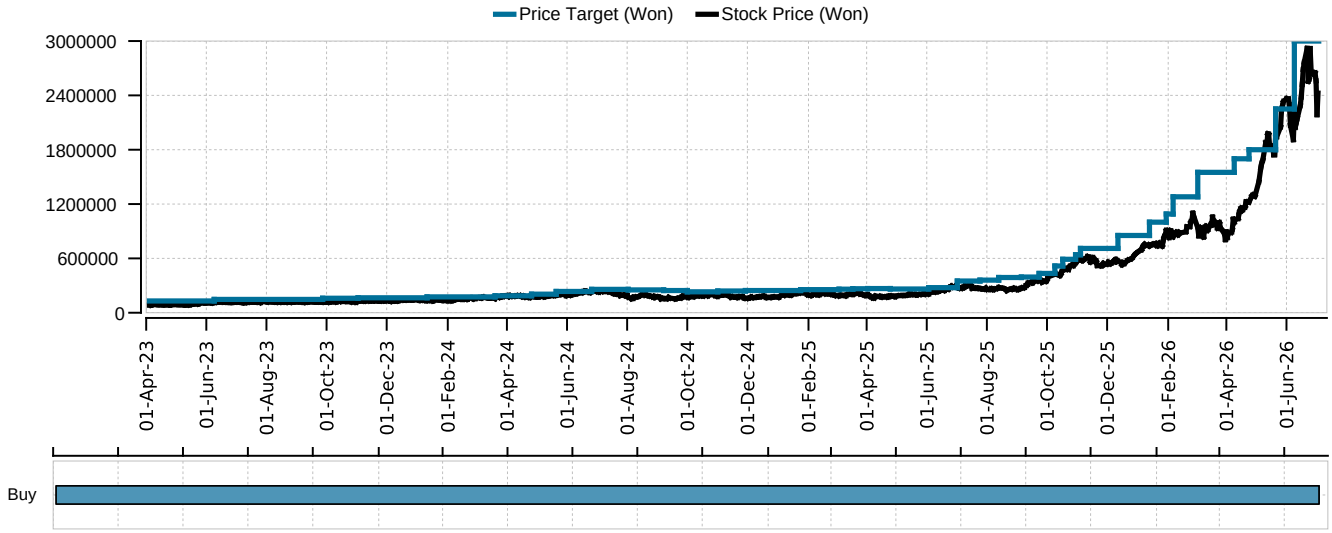
Samsung Electronics (Won)



Date	Stock Price (Won)	Price Target (Won)	Rating
2023-04-03	63100	80000	Buy
2023-04-07	65000	85000	Buy
2023-06-08	70900	91000	Buy
2023-09-26	68600	97000	Buy
2023-11-01	68600	100000	Buy
2024-01-10	73600	104000	Buy
2024-04-30	77500	107000	Buy
2024-05-20	78900	116000	Buy
2024-09-06	68900	104000	Buy
2024-09-30	61500	97000	Buy
2024-10-08	60300	95000	Buy
2024-10-31	59200	72000	Buy
2025-01-09	56100	66000	Buy
2025-01-31	52400	64000	Buy
2025-03-17	57600	66000	Buy
2025-04-23	55700	65000	Buy
2025-07-01	60200	65000	Neutral
2025-08-01	68900	72000	Neutral
2025-08-12	71100	75000	Neutral
2025-09-22	83500	85000	Neutral
2025-10-08	89000	93000	Neutral
2025-10-16	97700	112000	Buy
2025-10-31	107500	118000	Buy
2025-11-04	104900	128000	Buy
2025-12-11	107300	154000	Buy
2026-01-12	138800	172000	Buy
2026-01-30	160500	177000	Buy
2026-02-05	159300	242000	Buy
2026-03-02	216500	257000	Buy
2026-04-08	210500	266000	Buy
2026-04-30	220500	275000	Buy
2026-05-20	276000	400000	Buy
2026-06-08	295500	550000	Buy

Source: UBS Global Research; LSEG Eikon as of 03-Jul-2026. All prices as of local market close. Ratings as of date shown.

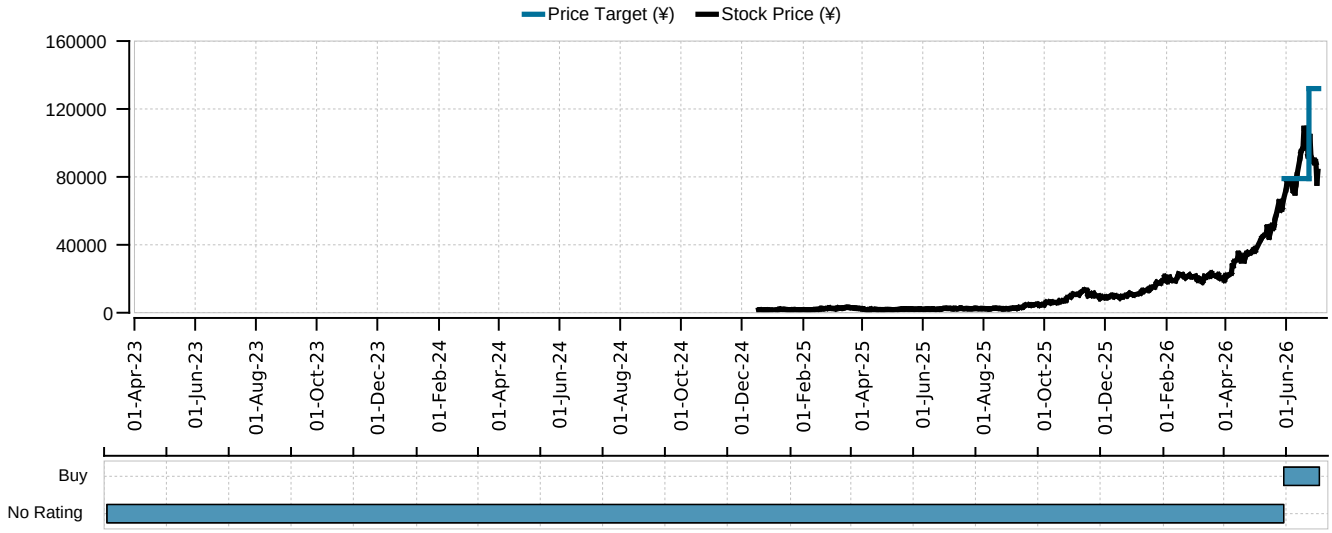
SK Hynix (Won)



Date	Stock Price (Won)	Price Target (Won)	Rating
2023-04-03	87200	130000	Buy
2023-06-08	109700	147000	Buy
2023-09-26	115000	160000	Buy
2023-11-01	120300	165000	Buy
2024-01-10	133500	175000	Buy
2024-03-19	160200	187000	Buy
2024-04-25	170600	205000	Buy
2024-05-20	190100	235000	Buy
2024-06-25	225000	258000	Buy
2024-08-01	193300	252000	Buy
2024-09-06	156400	245000	Buy
2024-09-30	174600	231000	Buy
2024-10-31	186300	240000	Buy
2024-11-28	161100	245000	Buy
2025-01-23	219500	255000	Buy
2025-03-03	190200	260000	Buy
2025-03-17	206000	265000	Buy
2025-04-01	197000	267000	Buy
2025-04-24	178300	262000	Buy
2025-06-02	207500	277000	Buy
2025-07-01	285500	350000	Buy
2025-07-24	269500	360000	Buy
2025-08-12	269000	389000	Buy
2025-09-04	265500	394000	Buy
2025-09-22	351000	434000	Buy
2025-10-08	395500	516000	Buy
2025-10-16	452500	590000	Buy
2025-10-29	558000	640000	Buy
2025-11-03	620000	710000	Buy
2025-12-11	565000	853000	Buy
2026-01-12	749000	1000000	Buy
2026-01-29	861000	1090000	Buy
2026-02-05	842000	1280000	Buy
2026-03-02	1061000	1550000	Buy
2026-04-08	1033000	1700000	Buy
2026-04-23	1225000	1800000	Buy
2026-05-20	1745000	2250000	Buy
2026-06-08	1911000	3000000	Buy

Source: UBS Global Research; LSEG Eikon as of 03-Jul-2026. All prices as of local market close. Ratings as of date shown.

Kioxia Holdings Corp (¥)



Date	Stock Price (¥)	Price Target (¥)	Rating
2023-04-03	NaN	-	No Rating
2026-05-29	65850	79000	Buy
2026-06-23	92290	132000	Buy

Source: UBS Global Research; LSEG Eikon as of 03-Jul-2026. All prices as of local market close. Ratings as of date shown.

Company profile and fee and risk statement under the Japanese Financial Instruments & Exchange Law

Company Name etc: UBS Securities Japan Co., Ltd., Financial Instruments & Exchange Firm, Kanto Local Financial Bureau (Kinsho) No.2633

Associated Memberships: Japan Securities Dealers' Association, the Financial Futures Association of Japan, and Type II Financial Instruments Firms Association

UBS Securities Japan Co., Ltd. will receive a brokerage fee based on an individual contract and no standard upper limit or calculating method. For the trading of domestic stocks, consumption tax is added to the fee. For the trading of foreign stock, fee on the foreign stock exchange or foreign tax may be charged in addition to the domestic fee and tax. Those amounts may vary depending on the jurisdiction. There is a risk that a loss may occur due to a change in the price of the stock in the case of trading stocks, and that a loss may occur due to the exchange rate in the case of trading foreign stocks. There is a risk that a loss may occur due to a change in the price or performance of the properties in the portfolio in the case of trading REITs.

UBS Securities Japan Co., Ltd. will only receive the purchasing amounts for trading unlisted bonds (JGBs, municipals, government guaranteed bonds, corporate bonds) when UBS Securities Japan Co., Ltd. is the counterparty. There is a risk that a loss may occur due to a change in the price of the bond caused by the fluctuations in the interest rates, and that a loss may occur due to the exchange rate in the case of trading foreign bonds.

For the detailed information of commissions and risks associated with individual financial instrument transactions, please carefully read the documents provided prior to the execution of the contracts or other materials for customers, etc. before executing any transaction. If you need any clarification, please consult with our sales person in charge.

The Disclaimer relevant to Global Wealth Management clients follows the Global Research Disclaimer. The Disclaimer relevant to Credit Suisse Wealth Management follows the Global Wealth Management Disclaimer.

UBS Global Research Disclaimer

This document has been prepared by UBS Securities Pte. Ltd., an affiliate of UBS AG. UBS AG, its subsidiaries, branches and affiliates, including former Credit Suisse AG and its subsidiaries, branches and affiliates are referred to herein as "UBS".

Any opinions expressed in this document may change without notice and are only current as of the date of publication. Different areas, groups, and personnel within UBS may produce and distribute separate research products independently of each other. For example, research publications from **UBS CIO** are produced by UBS Global Wealth Management. **UBS Global Research** is produced by UBS Investment Bank. Research methodologies and rating systems of each separate research organization may differ, for example, in terms of investment recommendations, investment horizon, model assumptions, and valuation methods. As a consequence, except for certain economic forecasts (for which UBS CIO and UBS Global Research may collaborate), investment recommendations, ratings, price targets, and valuations provided by each of the separate research organizations may be different, or inconsistent. You should refer to each relevant research product for the details as to their methodologies and rating system. Not all clients may have access to all products from every organization. Each research product is subject to the policies and procedures of the organization that produces it.

This document is provided solely to recipients who are expressly authorized by UBS to receive it. If you are not so authorized you must immediately destroy the document.

UBS Global Research is provided to our clients through UBS Neo, and in certain instances, UBS.com and any other system or distribution method specifically identified in one or more communications distributed through UBS Neo or UBS.com (each a system) as an approved means for distributing UBS Global Research. It may also be made available through third party vendors and distributed by UBS and/or third parties via e-mail or alternative electronic means.

All UBS Global Research is available on UBS Neo. Please contact your UBS sales representative if you wish to discuss your access to UBS Neo. Where UBS Global Research refers to "UBS Evidence Lab Inside" or has made use of data provided by UBS Evidence Lab and you would like to access that data please contact your UBS sales representative. UBS Evidence Lab data is available on UBS Neo. The level and types of services provided by UBS Global Research and UBS Evidence Lab to a client may vary depending upon various factors such as a client's individual preferences as to the frequency and manner of receiving communications, a client's risk profile and investment focus and perspective (e.g., market wide, sector specific, long-term, short-term, etc.), the size and scope of the overall client relationship with UBS Global Research and UBS Evidence Lab and legal and regulatory constraints. UBS HOLT and UBS Pharma Values are offerings of UBS Global Research. HOLT Lens is a corporate performance platform offering that provides an objective accounting-led framework for comparing and valuing companies and is available to clients of UBS Global Research; for further details and pricing please contact your UBS Sales representative. In particular, HOLT has a variety of warranted prices based on the scenario chosen; please mail UBS Securities LLC, 11 Madison Avenue, New York, NY 10010, USA, Attention: Investment Research, if you are interested in the warranted price on a particular company, again subject to commercial considerations. UBS Pharma Values is an analytical tool that involves the creation of a number of individual product net present value calculations, based on published forecasts of sales for pharmaceuticals, and is available to clients of UBS Global Research; for further details and pricing please contact your UBS Sales representative. For all other specific disclaimers, please see https://www.ubs.com/disclosures.

When you receive UBS Global Research through a system, your access and/or use of such UBS Global Research is subject to this UBS Global Research Disclaimer and to the UBS Neo Platform Use Agreement (the "Neo Terms") together with any other relevant terms of use governing the applicable System.

When you receive UBS Global Research via a third party vendor, e-mail or other electronic means, you agree that use shall be subject to this UBS Global Research Disclaimer, the Neo Terms and where applicable the UBS Investment Bank terms of business (https://www.ubs.com/global/en/investment-bank/regulatory.html) and to UBS's Terms of Use/Disclaimer (https://www.ubs.com/global/en/legalinfo2/disclaimer.html). In addition, you consent to UBS processing your personal data and using cookies in accordance with our Privacy Statement (https://www.ubs.com/global/en/legalinfo2/privacy.html) and cookie notice (https://www.ubs.com/global/en/legal/privacy/users.html).

If you receive UBS Global Research, whether through a System or by any other means, you agree that you shall not copy, revise, amend, create a derivative work, provide to any third party, or in any way commercially exploit any UBS research provided via UBS Global Research or otherwise, and that you shall not extract data from any research or estimates provided to you via UBS Global Research or otherwise, without the prior written consent of UBS. You agree not to use UBS Global Research in any artificial intelligence system, without the prior written consent of UBS.

In certain circumstances you may receive UBS Global Research otherwise than in the capacity of a client of UBS and you understand and agree that under these circumstances (i) the UBS Global Research is provided to you for information purposes only; (ii) for the purposes of receiving it you are not intended to be and will not be treated as a "client" of UBS for any legal or regulatory purpose; (iii) the UBS Global Research must not be relied on or acted upon for any purpose; and (iv) such content is subject to the relevant disclaimers that follow.

This document is for distribution only as may be permitted by law. It is not directed to, or intended for distribution to or use by, any person or entity who is a citizen or resident of or located in any locality, state, country or other jurisdiction where such distribution, publication, availability or use would be contrary to law or regulation or would subject UBS to any registration or licensing requirement within such jurisdiction.

This document is a general communication and is educational in nature; it is not an advertisement nor is it a solicitation or an offer to buy or sell any financial instruments or to participate in any particular trading strategy. Nothing in this document constitutes a representation that any investment strategy or recommendation is suitable or appropriate to an investor's individual circumstances or otherwise constitutes a personal recommendation. By providing this document, none of UBS or its representatives has any responsibility or authority to provide or have provided investment advice in a fiduciary capacity or otherwise. Investments involve risks, and investors should exercise prudence and their own judgment in making their investment decisions. None of UBS or its representatives is suggesting that the recipient or any other person take a specific course of action or any action at all. The recipient should carefully read this document in its entirety and not draw inferences or conclusions from the rating alone. By receiving this document, the recipient acknowledges and agrees with the intended purpose described above and further disclaims any expectation or belief that the information constitutes investment advice to the recipient or otherwise purports to meet the investment objectives of the recipient. The financial instruments described in the document may not be eligible for sale in all jurisdictions or to certain categories of investors.

Options, structured derivative products and futures (including OTC derivatives) are not suitable for all investors. Trading in these instruments is considered risky and may be appropriate only for sophisticated investors. Prior to buying or selling an option, and for the complete risks relating to options, you must receive a copy of "The Characteristics and Risks of Standardized Options." You may read the document at https://www.theocc.com/publications/risks/riskchap1.jsp or ask your salesperson for a copy. Various theoretical explanations of the risks associated with these instruments have been published. Supporting documentation for any claims, comparisons, recommendations, statistics or other technical data will be supplied upon request. Past performance is not necessarily indicative of future results. Transaction costs may be significant in option strategies calling for multiple purchases and sales of options, such as spreads and straddles. Because of the importance of tax considerations to many options transactions, the investor considering options should consult with his/her tax advisor as to how taxes affect the outcome of contemplated options transactions.

Mortgage and asset-backed securities may involve a high degree of risk and may be highly volatile in response to fluctuations in interest rates or other market conditions. Foreign currency rates of exchange may adversely affect the value, price or income of any security or related instrument referred to in the document. For investment advice, trade execution or other enquiries, clients should contact their local sales representative.

UBS notes that no globally accepted framework or definition (legal, regulatory or otherwise) currently exists, nor is there a market consensus as to what constitutes an "ESG" (Environmental, Social or Governance) or an equivalent-label, or as to what precise attributes are required for the Information (as defined below) to be defined as ESG or equivalently-labelled. Any information, data or other content including from a third party source contained, referred to herein or used for whatsoever purpose by UBS or a third party ("Information"), in relation to any actual or potential ESG objective, issue or consideration is not intended to be relied upon for ESG classification, regulatory regime or industry initiative purposes ("ESG Regimes"). Nothing in these materials is intended to convey, suggest or indicate that UBS considers or represents any product, service, person or body mentioned in these materials as meeting or qualifying for any ESG classification, labelling or similar standards that may exist under the ESG Regimes. UBS has not conducted any assessment of compliance with ESG Regimes. Parties are reminded to make their own assessments for these purposes.

The value of any investment or income may go down as well as up, and investors may not get back the full (or any) amount invested. Past performance is not necessarily a guide to future performance. Neither UBS nor any of its directors, employees or agents accepts any liability for any loss (including investment loss) or damage arising out of the use of all or any of the Information.

Prior to making any investment or financial decisions, any recipient of this document or the information should take steps to understand the risk and return of the investment and seek individualized advice from his or her personal financial, legal, tax and other professional advisors that takes into account all the particular facts and circumstances of his or her investment objectives.

Any prices stated in this document are for information purposes only and do not represent valuations for individual securities or other financial instruments. There is no representation that any transaction can or could have been effected at those prices, and any prices do not necessarily reflect UBS's internal books and records or theoretical model-based valuations and may be based on certain assumptions. Different assumptions by UBS or any other source may yield substantially different results.

No representation or warranty, either expressed or implied, is provided in relation to the accuracy, completeness or reliability of the information contained in any materials to which this document relates (the "Information"), except with respect to Information concerning UBS. The Information is not intended to be a complete statement or summary of the securities, markets or developments referred to in the document. UBS does not undertake to update or keep current the Information. Any statements contained in this report attributed to a third party represent UBS's interpretation of the data, information and/or opinions provided by that third party either publicly or through a subscription service, and such use and interpretation have not been reviewed by the third party. In no circumstances may this document or any of the Information (including any forecast, value, index or other calculated amount ("Values")) be used for any of the following purposes:

(i) valuation or accounting purposes;

(ii) to determine the amounts due or payable, the price or the value of any financial instrument or financial contract; or

(iii) to measure the performance of any financial instrument including, without limitation, for the purpose of tracking the return or performance of any Value or of defining the asset allocation of portfolio or of computing performance fees.

By receiving this document and the Information you will be deemed to represent and warrant to UBS that you will not use this document or any of the Information for any of the above purposes or otherwise rely upon this document or any of the Information.

UBS has policies and procedures, which include, without limitation, independence policies and permanent information barriers, that are intended, and upon which UBS relies, to manage potential conflicts of interest and control the flow of information within divisions of UBS and among its subsidiaries, branches and affiliates. UBS does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. For further information on the ways in which UBS Global Research manages conflicts and maintains independence of its research products, historical performance information and certain additional disclosures concerning UBS Global Research recommendations, please visit https://www.ubs.com/disclosures.

UBS Global Research will initiate, update and cease coverage solely at the discretion of UBS Global Research Management, which will also have sole discretion on the timing and frequency of any published research product. The analysis contained in this document is based on numerous assumptions. All material information in relation to published research reports, such as valuation methodology, risk statements, underlying assumptions (including sensitivity analysis of those assumptions), ratings history etc. as required by the Market Abuse Regulation, can be found on UBS Neo. Different assumptions could result in materially different results.

UBS Global Research may utilise artificial intelligence tools ("AI Tools") in the preparation of this document. Notwithstanding any such use of AI Tools, this document has undergone human review.

The analyst(s) responsible for the preparation of this document may interact with trading desk personnel, sales personnel and other parties for the purpose of gathering, applying and interpreting market information. UBS relies on information barriers to control the flow of information contained in one or more areas within UBS into other areas, units, groups or affiliates of UBS. The compensation of the analyst who prepared this document is determined exclusively by UBS Global Research management and senior management (not including investment banking). Analyst compensation is not based on investment banking revenues; however, compensation may relate to the revenues of UBS and/or its divisions as a whole, of which investment banking, sales and trading are a part, and UBS as a whole.

For financial instruments admitted to trading on an EU regulated market: UBS (excluding UBS Securities LLC) acts as a market maker or liquidity provider (in accordance with the interpretation of these terms under English law or, if not carried out by UBS in the UK the law of the relevant jurisdiction in which UBS determines it carries out the activity) in the financial instruments of the issuer save that where the activity of liquidity provider is carried out in accordance with the definition given to it by the laws and regulations of any other EU jurisdictions, such information is separately disclosed in this document. For financial instruments admitted to trading on a non-EU regulated market: UBS may act as a market maker save that where this activity is carried out in the US in accordance with the definition given to it by the relevant laws and regulations, such activity will be specifically disclosed in this document. UBS may have issued a warrant the value of which is based on one or more of the financial instruments referred to in the document. UBS and its affiliates and employees may have long or short positions, trade as principal and buy and sell in instruments or derivatives identified herein; such transactions or positions may be inconsistent with the opinions expressed in this document.

Within the past 12 months UBS may have received or provided investment services and activities or ancillary services as per MiFID II which may have given rise to a payment or promise of a payment in relation to these services from or to this company.

Please note that all transactions conducted by UBS are consistent with sanctions regulations imposed by Switzerland, the European Union, the United Nations, the United Kingdom and the United States, per UBS' global sanctions policy. UBS opinion as to future investment worthiness assumes no new sanctions are imposed.

US persons are prohibited from purchasing or selling securities of certain companies designated as being associated with the Chinese Military in accordance with the amended US Presidential Executive Order 13959.

United Kingdom: This material is distributed by UBS AG, London Branch to persons who are eligible counterparties or professional clients. UBS AG, London Branch is authorised by the Prudential Regulation Authority and subject to regulation by the Financial Conduct Authority and limited regulation by the Prudential Regulation Authority. **Europe:** Except as otherwise specified herein, these materials are distributed by UBS Europe SE, a subsidiary of UBS AG, to persons who are eligible counterparties or professional clients (as detailed in the Bundesanstalt fur Finanzdienstleistungsaufsicht (BaFin) Rules and according to MIFID) and are only available to such persons. The information does not apply to, and should not be relied upon by, retail clients. UBS Europe SE is authorised by the European Central Bank (ECB) and regulated by the BaFin and the ECB. **Germany, Luxembourg, the Netherlands, Belgium and Ireland:** Where an analyst of UBS Europe SE has contributed to this document, the document is also deemed to have been prepared by UBS Europe SE. In all cases it is distributed by UBS Europe SE and UBS AG, London Branch. **Turkey:** Distributed by UBS AG, London Branch. No information in this document is provided for the purpose of offering, marketing and sale by any means of any capital market instruments and services in the Republic of Turkey. Therefore, this document may not be considered as an offer made or to be made to residents of the Republic of Turkey. UBS AG, London Branch is not licensed by the Turkish Capital Market Board under the provisions of the Capital Market Law (Law No. 6362). Accordingly, neither this document nor any other offering material related to the instruments/services may be utilized in connection with providing any capital market services to persons within the Republic of Turkey without the prior approval of the Capital Market Board. However, according to article 15 (d) (ii) of the Decree No. 32, there is no restriction on the purchase or sale of the securities abroad by residents of the Republic of Turkey. **Poland:** Distributed by UBS Europe SE (spolka z ograniczona odpowiedzialnoscia) Oddzial w Polsce. Where an analyst of UBS Europe SE (spolka z ograniczona odpowiedzialnoscia) Oddzial w Polsce has contributed to this document, the document is also deemed to have been prepared by UBS Europe SE (spolka z ograniczona odpowiedzialnoscia) Oddzial w Polsce. **Russia:** Prepared and distributed by UBS Bank (OOO). Should not be construed as an individual Investment Recommendation for the purpose of the Russian Law - Federal Law #39-FZ ON THE SECURITIES MARKET Articles 6.1-6.2.**Switzerland:** Distributed by UBS AG to persons who are institutional investors only. UBS AG is regulated by the Swiss Financial Market Supervisory Authority (FINMA). **Italy:** Prepared by UBS Europe SE and distributed by UBS Europe SE and UBS Europe SE, Italy Branch. Where an analyst of UBS Europe SE, Italy Branch has contributed to this document, the document is also deemed to have been prepared by UBS Europe SE, Italy Branch. **France:** Prepared by UBS Europe SE and distributed by UBS Europe SE and UBS Europe SE, France Branch. Where an analyst of UBS Europe SE, France Branch has contributed to this document, the document is also deemed to have been prepared by UBS Europe SE, France Branch. **Spain:** Prepared by UBS Europe SE and distributed by UBS Europe SE and UBS Europe SE, Spain Branch. Where an analyst of UBS Europe SE, Spain Branch has contributed to this document, the document is also deemed to have been prepared by UBS Europe SE, Spain Branch. **Sweden:** Prepared by UBS Europe SE and distributed by UBS Europe SE and UBS Europe SE, Sweden Branch. Where an analyst of UBS Europe SE, Sweden Branch has contributed to this document, the document is also deemed to have been prepared by UBS Europe SE, Sweden Branch. **South Africa:** Distributed by UBS South Africa (Pty) Limited (Registration No. 1995/011140/07), an authorised user of the JSE and an authorised Financial Services Provider (FSP 7328). **Saudi Arabia:** This document has been issued by UBS AG (and/or any of its subsidiaries, branches or affiliates), a public company limited by shares, incorporated in Switzerland with its registered offices at Aeschenvorstadt 1, CH-4051 Basel and Bahnhofstrasse 45, CH-8001 Zurich. This publication has been approved by UBS Saudi Arabia (a subsidiary of UBS AG), a Saudi closed joint stock company incorporated in the Kingdom of Saudi Arabia under commercial register number 1010257812 having its registered office at Tatweer Towers, P.O. Box 75724, Riyadh 11588, Kingdom of Saudi Arabia. UBS Saudi Arabia is authorized and regulated by the Capital Market Authority to conduct securities business under license number 08113-37. **UAE / Dubai:** The information distributed by UBS AG Dubai Branch is only intended for Professional Clients and/or Market Counterparties, as classified under the DFSA rulebook. No other person should act upon this material/communication. The information is not for further distribution within the United Arab Emirates. UBS AG Dubai Branch is regulated by the DFSA in the DIFC. UBS is not licensed to provide banking services in the UAE by the Central Bank of the UAE, nor is it licensed by the UAE Securities and Commodities Authority. **Israel:** This Material is distributed by UBS AG, London Branch. UBS Securities Israel Ltd is a licensed Investment Marketer that is supervised by the Israel Securities Authority (ISA). UBS AG, London Branch and its affiliates incorporated outside Israel are not licensed under the Israeli Advisory Law. UBS may engage among others in issuance of Financial Assets or in distribution of Financial Assets of other issuers for fees or other benefits. UBS AG, London Branch and its affiliates may prefer various Financial Assets to which they have or may have an Affiliation (as such term is defined under the Israeli Advisory Law). Nothing in this Material should be considered as investment advice under the Israeli Advisory Law. This Material is being issued only to and/or is directed only at persons who are Eligible Clients within the meaning of the Israeli Advisory Law, and this Material must not be furnished to, relied on or acted upon by any other persons. **United States:** Distributed to US persons by either UBS Securities LLC or by UBS Financial Services Inc., subsidiaries of UBS AG; or by a group, subsidiary or affiliate of UBS AG that is not registered as a US broker-dealer (a **'non-US affiliate'**) to major US institutional investors only. UBS Securities LLC or UBS Financial Services Inc. accepts responsibility for the content of a report prepared by another non-US affiliate when distributed to US persons by UBS Securities LLC or UBS Financial Services Inc. All transactions by a US person in the securities mentioned in this report must be effected through UBS Securities LLC or UBS Financial Services Inc., and not through a non-US affiliate. UBS Securities LLC is not acting as a municipal advisor to any municipal entity or obligated person within the meaning of

Section 15B of the Securities Exchange Act (the "Municipal Advisor Rule"), and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of the Municipal Advisor Rule. **Canada:** Distributed by UBS Securities Canada Inc., a registered investment dealer in Canada and a Member-Canadian Investor Protection Fund, or by another affiliate of UBS AG that is registered to conduct business in Canada or is otherwise exempt from registration. **Brazil:** Except as otherwise specified herein, this Material is prepared by UBS BB Corretora de Câmbio, Títulos e Valores Mobiliários S.A. (UBS BB CCTVM) to persons who are eligible investors residing in Brazil, which are considered to be Professional Investors (Investidores Profissionais), as designated by the applicable regulation, mainly the CVM Resolution No. 30 from the 11th of May 2021 (determines the duty to verify the suitability of products, services and transactions with regards to the client´s profile). UBS BB CCTVM is a subsidiary of UBS BB Servicos de Assessoria Financeira e Participacoes S.A. ("UBS BB"). UBS BB is an association between UBS AG and Banco do Brasil (through its subsidiary BB – Banco de Investimentos S.A.), of which UBS AG is the majority owner and which provides investment banking services and coverage in Brazil, Argentina, Chile, Paraguay, Peru and Uruguay. UBS BB CCTVM is regulated by the Comissao de Valores Mobiliarios (CVM) and by the Central Bank of Brazil. Ombudsman: 0800-940-0266/ https://www.ubs.com/br/pt/ubsbb-investment-bank/ombudsman.html. UBS may hold relevant financial and commercial interest in relation to the company subject to this Research report. **Hong Kong:** Distributed by UBS Securities Asia Limited. Please contact local licensed persons of UBS Securities Asia Limited in respect of any matters arising from, or in connection with, the analysis or document **Singapore:** Distributed by UBS Securities Pte. Ltd. [Co. Reg. No.: 198500648C] or UBS AG, Singapore Branch. Please contact UBS Securities Pte. Ltd., an exempt financial adviser under the Singapore Financial Advisers Act (Cap. 110); or UBS AG, Singapore Branch, an exempt financial adviser under the Singapore Financial Advisers Act (Cap. 110) and a wholesale bank licensed under the Singapore Banking Act (Cap. 19) regulated by the Monetary Authority of Singapore, in respect of any matters arising from, or in connection with, the analysis or document. The recipients of this document represent and warrant that they are accredited and institutional investors as defined in the Securities and Futures Act (Cap. 289). **Japan:** Distributed by UBS Securities Japan Co., Ltd. to professional investors (except as otherwise permitted). Where this report has been prepared by UBS Securities Japan Co., Ltd., UBS Securities Japan Co., Ltd. is the author, publisher and distributor of the report. Distributed by UBS AG, Tokyo Branch to Professional Investors (except as otherwise permitted) in relation to foreign exchange and other banking businesses when relevant. **Australia:** Clients of UBS AG: Distributed by UBS AG (ABN 47 088 129 613 and holder of Australian Financial Services License No. 231087). For all other recipients: Distributed by UBS Securities Australia Ltd (ABN 62 008 586 481 and holder of Australian Financial Services License No. 231098). This document contains general information and/or general advice only and does not constitute personal financial product advice. As such, the Information in this document has been prepared without taking into account any investor's objectives, financial situation or needs, and investors should, before acting on the Information, consider the appropriateness of the Information, having regard to their objectives, financial situation and needs. If the Information contained in this document relates to the acquisition, or potential acquisition of a particular financial product by a 'Retail' client as defined by section 761G of the Corporations Act 2001 where a Product Disclosure Statement would be required, the retail client should obtain and consider the Product Disclosure Statement relating to the product before making any decision about whether to acquire the product. **New Zealand:** Distributed by UBS New Zealand Ltd. UBS New Zealand Ltd is not a registered bank in New Zealand. You are being provided with this publication or material because you have indicated to UBS that you are a "wholesale client" within the meaning of clause 4 of schedule 5 of the Financial Markets Conduct Act 2013 of New Zealand (Permitted Client). This publication or material is not intended for clients who are not Permitted Clients (non-permitted Clients). If you are a non-permitted Client you must not rely on this publication or material. If despite this warning you nevertheless rely on this publication or material, you hereby (i) acknowledge that you may not rely on the content of this publication or material and that any recommendations or opinions in such this publication or material are not made or provided to you, and (ii) to the maximum extent permitted by law (a) indemnify UBS and its associates or related entities (and their respective Directors, officers, agents and Advisors) (each a 'Relevant Person') for any loss, damage, liability or claim any of them may incur or suffer as a result of, or in connection with, your unauthorised reliance on this publication or material and (b) waive any rights or remedies you may have against any Relevant Person for (or in respect of) any loss, damage, liability or claim you may incur or suffer as a result of, or in connection with, your unauthorised reliance on this publication or material. **Korea:** Distributed in Korea by UBS Securities Pte. Ltd., Seoul Branch. This report may have been edited or contributed to from time to time by affiliates of UBS Securities Pte. Ltd., Seoul Branch. This material is intended for professional/institutional clients only and not for distribution to any retail clients. **Malaysia:** This material is authorized to be distributed in Malaysia by UBS Securities Malaysia Sdn. Bhd (Capital Markets Services License No.: CMSL/A0063/2007). This material is intended for professional/institutional clients only and not for distribution to any retail clients. **India:** Distributed by UBS Securities India Private Ltd. (Corporate Identity Number U67120MH1996PTC097299) 2/F, 3 North Avenue, Maker Maxity, Bandra Kurla Complex, Bandra (East), Mumbai (India) 400051. Phone: +912261556000. It provides brokerage services bearing SEBI Registration Number: INZ000259830; Merchant Banking services bearing SEBI Registration Number: INM000013101; and Research Analyst services bearing SEBI Registration Number: INH000001204. Name of Compliance Officer Mr. Parameshwaran Shivaramakrishnan, Phone : +912261556151, Email : parameshwaran.s@ubs.com, Name of Grievance Officer Parameshwaran Shivaramakrishnan, Phone : +912261556151, Email: ol-ubs-sec-compliance@ubs.com Registration granted by SEBI, and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. UBS may have debt holdings or positions in the subject Indian company/companies. UBS may have financial interests (e.g. loan/derivative products, rights to or interests in investments, etc.) in the subject Indian company / companies from time to time. Within the past 12 months, UBS may have received compensation for non-investment banking securities-related services and/or non-securities services from the subject Indian company/companies. The subject company/companies may have been a client/clients of UBS during the 12 months preceding the date of distribution of the research report with respect to investment banking and/or non-investment banking securities-related services and/or non-securities services. With regard to information on associates, please refer to the Annual Report at: https://www.ubs.com/global/en/about_ubs/investor_relations/annualreporting.html The Research Annual Compliance Report for UBS Securities India Private Limited is available on www.ubs.com/ubssi under Research tab. **Taiwan:** Except as otherwise specified herein, this material may not be distributed in Taiwan. Information and material on securities/instruments that are traded in a Taiwan organized exchange is deemed to be issued and distributed by UBS Securities Pte. LTD., Taipei Branch, which is licensed and regulated by Taiwan Financial Supervisory Commission. Save for securities/instruments that are traded in a Taiwan organized exchange, this material should not constitute "recommendation" to clients or recipients in Taiwan for the covered companies or any companies mentioned in this document. No portion of the document may be reproduced or quoted by the press or any other person without authorisation from UBS. **Indonesia:** This report is being distributed by PT UBS Sekuritas Indonesia and is delivered by its licensed employee(s), including marketing/sales person, to its client. PT UBS Sekuritas Indonesia, having its registered office at Sequis Tower Level 22 unit 22-1,Jl.Jend. Sudirman, kav.71, SCBD lot 11B, Jakarta 12190. Indonesia, is a subsidiary company of UBS AG and licensed under Capital Market Law no. 8 year 1995, a holder of broker-dealer and underwriter licenses issued by the Capital Market and Financial Institution Supervisory Agency (now Otoritas Jasa Keuangan/OJK). PT UBS Sekuritas Indonesia is also a member of Indonesia Stock Exchange and supervised by Otoritas Jasa Keuangan (OJK). Neither this report nor any copy hereof may be distributed in Indonesia or to any Indonesian citizens except in compliance with applicable Indonesian capital market laws and regulations. This report is not an offer of securities in Indonesia and may not be distributed within the territory of the Republic of Indonesia or to Indonesian citizens in circumstance which constitutes an offering within the meaning of Indonesian capital market laws and regulations.

The disclosures contained in research documents produced by UBS AG, London Branch or UBS Europe SE shall be governed by and construed in accordance with English law.

UBS specifically prohibits the redistribution of this document in whole or in part without the written permission of UBS and in any event UBS accepts no liability whatsoever for any redistribution of this document or its contents or the actions of third parties in this respect. Images may depict objects or elements that are protected by third party copyright, trademarks and other intellectual property rights. © UBS 2026. The key symbol and UBS are among the registered and unregistered trademarks of UBS. All rights reserved.

Global Wealth Management Disclaimer

You receive this document in your capacity as a client of UBS Global Wealth Management. This publication has been distributed to you by UBS Switzerland AG (regulated by FINMA in Switzerland) or its affiliates (**"UBS"**) with whom you have a banking relationship with. The full name of the distributing affiliate and its competent authority can be found in the country-specific disclaimer at the end of this document. UBS may utilise artificial intelligence tools ("AI Tools") in the preparation of this document. Notwithstanding any such use of AI Tools, this document has undergone human review.

The date and time of the first dissemination of this publication is the same as the date and time of its publication.

Risk information:

You agree that you shall not copy, revise, amend, create a derivative work, provide to any third party, or in any way commercially exploit any UBS research, and that you shall not extract data from any research or estimates, without the prior written consent of UBS.

This document is for distribution only as may be permitted by law. It is not directed to, or intended for distribution to or use by, any person or entity who is a citizen or resident of or located in any locality, state, country or other jurisdiction where such distribution, publication, availability or use would be contrary to law or regulation or would subject UBS to any registration or licensing requirement within such jurisdiction.

This document is **for your information only**; it is not an advertisement nor is it a solicitation or an offer to buy or sell any financial instruments or to participate in any particular trading strategy. Nothing in this document constitutes a representation that any investment strategy or recommendation is suitable or appropriate to an investor's individual circumstances or otherwise constitutes a personal recommendation. By providing this document, none of UBS or its representatives has any responsibility or authority to provide or have provided investment advice in a fiduciary capacity or otherwise. Investments involve risks, and investors should exercise prudence and their own judgment in making their investment decisions. None of UBS or its representatives is suggesting that the recipient or any other person take a specific course of action or any action at all. By receiving this document, the recipient acknowledges and agrees with the intended purpose described above and further disclaims any expectation or belief that the information constitutes investment advice to the recipient or otherwise purports to meet the investment objectives of the recipient. The financial instruments described in the document may not be eligible for sale in all jurisdictions or to certain categories of investors.

Options, derivative products and futures are not suitable for all investors, and trading in these instruments is considered risky. Mortgage and asset-backed securities may involve a high degree of risk and may be highly volatile in response to fluctuations in interest rates or other market conditions. Foreign currency rates of exchange may adversely affect the value, price or income of any security or related instrument referred to in the document. For investment advice, trade execution or other enquiries, clients should contact their local sales representative.

The value of any investment or income may go down as well as up, and investors may not get back the full (or any) amount invested. Past performance is not necessarily a guide to future performance. Neither UBS nor any of its directors, employees or agents accepts any liability for any loss (including investment loss) or damage arising out of the use of all or any of the information (as defined below).

Prior to making any investment or financial decisions, any recipient of this document or the information should take steps to understand the risk and return of the investment and seek individualized advice from his or her personal financial, legal, tax and other professional advisors that takes into account all the particular facts and circumstances of his or her investment objectives.

Any prices stated in this document are for information purposes only and do not represent valuations for individual securities or other financial instruments. There is no representation that any transaction can or could have been effected at those prices, and any prices do not necessarily reflect UBS's internal books and records or theoretical model-based valuations and may be based on certain assumptions. Different assumptions by UBS or any other source may yield substantially different results.

No representation or warranty, either expressed or implied, is provided in relation to the accuracy, completeness or reliability of the information contained in any materials to which this document relates (the "Information"), except with respect to Information concerning UBS. The Information is not intended to be a complete statement or summary of the securities, markets or developments referred to in the document. UBS does not undertake to update or keep current the Information. Any opinions expressed in this document may change without notice and may differ or be contrary to opinions expressed by other business areas or groups, personnel or other representative of UBS. Any statements contained in this report attributed to a third party represent UBS's interpretation of the data, information and/or opinions provided by that third party either publicly or through a subscription service, and such use and interpretation have not been reviewed by the third party. In no circumstances may this document or any of the Information (including any forecast, value, index or other calculated amount ("Values")) be used for any of the following purposes: (i) valuation or accounting purposes; (ii) to determine the amounts due or payable, the price or the value of any financial instrument or financial contract; or (iii) to measure the performance of any financial instrument including, without limitation, for the purpose of tracking the return or performance of any Value or of defining the asset allocation of portfolio or of computing performance fees.

By receiving this document and the Information you will be deemed to represent and warrant to UBS that you will not use this document or any of the Information for any of the above purposes or otherwise rely upon this document or any of the Information.

UBS has policies and procedures, which include, without limitation, independence policies and permanent information barriers, that are intended, and upon which UBS relies, to manage potential conflicts of interest and control the flow of information within divisions of UBS (including between Global Wealth Management and UBS Global Research) and among its subsidiaries, branches and affiliates. For further information on the ways in which UBS manages conflicts and maintains independence of its research products, historical performance information and certain additional disclosures concerning UBS research recommendations, please visit https://www.ubs.com/research-methodology.

Research will initiate, update and cease coverage solely at the discretion of research management, which will also have sole discretion on the timing and frequency of any published research product. The analysis contained in this document is based on numerous assumptions. Different assumptions could result in materially different results.

The analyst(s) responsible for the preparation of this document may interact with trading desk personnel, sales personnel and other parties for the purpose of gathering, applying and interpreting market information. UBS relies on information barriers to control the flow of information contained in one or more areas within UBS into other areas, units, groups or affiliates of UBS. The compensation of the analyst who prepared this document is determined exclusively by research management and senior management (not including investment banking). Analyst compensation is not based on investment banking revenues; however, compensation may relate to the revenues of UBS and/or its divisions as a whole, of which investment banking, sales and trading are a part, and UBS's subsidiaries, branches and affiliates as a whole.

For financial instruments admitted to trading on an EU regulated market: UBS AG, its affiliates or subsidiaries (excluding UBS Securities LLC) acts as a market maker or liquidity provider (in accordance with the interpretation of these terms in the UK) in the financial instruments of the issuer save that where the activity of liquidity provider is carried out in accordance with the definition given to it by the laws and regulations of any other EU jurisdictions, such information is separately disclosed in this document. For financial instruments admitted to trading on a non-EU regulated market: UBS may act as a market maker save that where this activity is carried out in the US in accordance with the definition given to it by the relevant laws and regulations, such activity will be specifically disclosed in this document. UBS may have issued a warrant the value of which is based on one or more of the financial instruments referred to in the document. UBS and its affiliates and employees may have long or short positions, trade as principal and buy and sell in instruments or derivatives identified herein; such transactions or positions may be inconsistent with the opinions expressed in this document.

Options and futures are not suitable for all investors, and trading in these instruments is considered risky and may be appropriate only for sophisticated investors. Prior to buying or selling an option, and for the complete risks relating to options, you must receive a copy of "Characteristics and Risks of Standardized Options". You may read the document at https://www.theocc.com/about/publications/character-risks.jsp or ask your financial advisor for a copy.

Investing in structured investments involves significant risks. For a detailed discussion of the risks involved in investing in any particular structured investment, you must read the relevant offering materials for that investment. Structured investments are unsecured obligations of a particular issuer with returns linked to the performance of an underlying asset. Depending on the terms of the investment, investors could lose all or a substantial portion of their investment based on the performance of the underlying asset. Investors could also lose their entire investment if the issuer becomes insolvent. UBS does not guarantee in any way the obligations or the financial condition of any issuer or the accuracy of any financial information provided by any issuer. Structured investments are not traditional investments and investing in a structured investment is not equivalent to investing directly in the underlying asset. Structured investments may have limited or no liquidity, and investors should be prepared to hold their investment to maturity. The return of structured investments may be limited by a maximum gain, participation rate or other feature. Structured investments may include call features and, if a structured investment is called early, investors would not earn any further return and may not be able to reinvest in similar investments with similar terms. Structured investments include costs and fees which are generally embedded in the price of the investment. The tax treatment of a structured investment may be complex and may differ from a direct investment in the underlying asset. UBS and its employees do not provide tax advice. Investors should consult their own tax advisor about their own tax situation before investing in any securities.

Important Information About Sustainable Investing Strategies: Sustainable investing strategies aim to consider and incorporate environmental, social and governance (ESG) factors into investment process and portfolio construction. Strategies across geographies approach ESG analysis and incorporate the findings in a variety of ways. Incorporating ESG factors or Sustainable Investing considerations may inhibit the portfolio manager's ability to participate in certain investment opportunities that otherwise would be consistent with its investment objective and other principal investment strategies. The returns on a portfolio incorporating ESG factors or Sustainable Investing considerations may be lower or higher than portfolios where ESG factors, exclusions, or other sustainability issues are not considered by the portfolio manager, and the investment opportunities available to such portfolios may differ.

Within the past 12 months UBS Switzerland AG, its affiliates or subsidiaries may have received or provided investment services and activities or ancillary services as per MiFID II which may have given rise to a payment or promise of a payment in relation to these services from or to this company.

Disclosures: If you require detailed information on disclosures of interest or conflict of interest as required by Market Abuse Regulation please contact the mailbox MAR_disclosure_twopager@ubs.com. Please note that e-mail communication is unsecured.

External Asset Managers / External Financial Consultants: In case this research or publication is provided to an External Asset Manager or an External Financial Consultant, UBS expressly prohibits that it is redistributed by the External Asset Manager or the External Financial Consultant and is made available to their clients and/or third parties.

Argentina: All securities that will be offered to you by UBS are not authorized by the Argentine Securities and Exchange Commission (CNV) and are not subject to its reporting, periodic information requirements, or oversight. Furthermore, the CNV has not reviewed or endorsed the information provided in any securities offering document, nor the accuracy of any accounting, financial, economic data, or any other information disclosed therein, which remains the sole responsibility of the respective issuer and the other parties involved. **Australia:** This document is provided by UBS Switzerland AG. UBS Switzerland AG does not hold an Australian Financial Services Licence (AFSL) and relies on an exemption to provide financial services to persons in Australia. This document is intended only for distribution to wholesale clients under the Corporations Act 2001 (Cth). UBS Switzerland AG is a related body corporate of UBS AG, Australia Branch and UBS Securities Australia Limited. This document may be distributed to clients by those entities, but it is provided by UBS Switzerland AG and is not provided under any of the other entities' AFSL. The information in this document is general in nature and is not intended to address the objectives, financial situation or needs of any particular individual or entity. Each recipient should consider their own objectives, financial situation or needs before acting on the advice and obtain the relevant Product Disclosure Statement (if required) before making any decision whether to acquire any product. In Australia, UBS entities, other than UBS AG, Australia Branch, are not authorized deposit-taking institutions for the purposes of the Banking Act 1959 (Cth.) and their obligations do not represent deposits or other liabilities of UBS AG, Australia Branch. UBS AG, Australia Branch does not guarantee or otherwise provide assurance in respect of the obligations of such UBS entities or the funds. An investor is exposed to investment risk including possible delays in repayment and loss of income and principal invested, as relevant. If you do not wish to receive marketing materials from UBS, please contact your UBS representative or the contact details listed in the Australia Privacy Notice: https://www.ubs.com/global/en/legal/privacy.html. Your personal data will be processed in accordance with this notice. **Bahrain**: This report is being

distributed by UBS AG, Bahrain Branch, duly licensed and regulated by the Central Bank of Bahrain (CBB) as an Investment Business Firm - Category 2 (Branch). Related financial services or products are only made available to Accredited Investors, as defined by the CBB, and are not intended for any other persons. UBS AG, Bahrain Branch is a Foreign Branch of UBS AG, Zurich/Switzerland and is located on Level 21, East Tower, Bahrain World Trade Centre, Manama, Kingdom of Bahrain. **Brazil**: This report is distributed in Brazil by UBS BB Corretora de Câmbio, Títulos e Valores Mobiliários S.A. or its affiliates ("UBS"). Pursuant to CVM Resolution No. 20/2021, of February 25, 2021, the author(s) of the report hereby certify(ies) that the views expressed in this report solely and exclusively reflect their personal opinions and have been prepared independently, including with respect to UBS. Part of the author(s)'s compensation is based on various factors, including the total revenues of the relevant UBS Group entity of which they are in employment of, but no part of the compensation has been, is, or will be related to the specific recommendations or views expressed in this report. In addition, UBS declares that: UBS has provided, and/or may in the future provide investment banking, brokerage, asset management, commercial banking and other financial services to the subject company/companies or its affiliates, for which they have received or may receive customary fees and commissions, and which constituted or may constitute relevant financial or commercial interests in relation to the subject company/companies or the subject securities. **Canada**: The information contained herein is not, and under no circumstances is to be construed as, a prospectus, an advertisement, a public offering, an offer to sell securities described herein, solicitation of an offer to buy securities described herein, in Canada or any province or territory thereof. Any offer or sale of the securities described herein in Canada will be made only under an exemption from the requirements to file a prospectus with the relevant Canadian securities regulators and only by a dealer properly registered under applicable securities laws or, alternatively, pursuant to an exemption from the dealer registration requirement in the relevant province or territory of Canada in which such offer or sale is made. Under no circumstances is the information contained herein to be construed as investment advice in any province or territory of Canada and is not tailored to the needs of the recipient. To the extent that the information contained herein references securities of an issuer incorporated, formed or created under the laws of Canada or a province or territory of Canada, any trades in such securities must be conducted through a dealer registered in Canada or, alternatively, pursuant to a dealer registration exemption. No securities commission or similar regulatory authority in Canada has reviewed or in any way passed upon these materials, the information contained herein or the merits of the securities described herein and any representation to the contrary is an offence. In Canada, this publication is distributed by UBS Investment Management Canada Inc. **China**: This report and any offering material such as term sheet, research report, other product or service documentation or any other information (the "Material") sent with this report was done so as a result of a request received by UBS from you and/or persons entitled to make the request on your behalf. Should you have received the material erroneously, UBS asks that you kindly delete it and inform UBS immediately. This report is prepared by UBS Switzerland AG or its offshore subsidiary or affiliate (collectively as "UBS Offshore"). UBS Offshore is an entity incorporated out of China and is not licensed, supervised or regulated in China to carry out banking or securities business. The recipient should not contact the analysts or UBS Offshore which produced this report for advice as they are not licensed to provide securities investment advice in China. UBS Investment Bank (including Research) has its own wholly independent research and views which at times may vary from the views of UBS Global Wealth Management. The recipient should not use this document or otherwise rely on any of the information contained in this report in making investment decisions and UBS takes no responsibility in this regard. **Czech Republic**: UBS is not a licensed bank in the Czech Republic and thus is not allowed to provide regulated banking or investment services in the Czech Republic. This communication and/or material is distributed for marketing purposes and constitutes a "Commercial Message" under the laws of Czech Republic in relation to banking and/or investment services. Please notify UBS if you do not wish to receive any further correspondence. **Denmark**: This publication is not intended to constitute a public offer under Danish law. It is distributed only for information purposes to clients of UBS Europe SE, Denmark Branch, filial af UBS Europe SE, with place of business at Sankt Annae Plads 13, 1250 Copenhagen, Denmark, registered with the Danish Commerce and Companies Agency, under No. 38 17 24 33. UBS Europe SE, Denmark Branch, filial af UBS Europe SE is subject to the joint supervision of the European Central Bank, the German Central Bank (Deutsche Bundesbank), the German Federal Financial Services Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, "BaFin"), as well as of the Danish Financial Supervisory Authority (Finanstilsynet), to which this publication has not been submitted for approval. UBS Europe SE is a credit institution constituted under German law in the form of a Societas Europaea, duly authorized by BaFin. **Egypt**: Securities or other investment products are not being offered or sold by UBS to the public in Egypt and they have not been and will not be registered with the Egyptian Financial Regulatory Authority (FRA). **France**: This publication is not intended to constitute a public offer under French law, it does not constitute a personal recommendation as it is distributed only for information purposes to clients of UBS Europe SE Succursale de France (a branch of UBS Europe SE), having its registered office at 39, rue du Colisée, 75008 Paris, France, registered with the "Registre du Commerce et des Sociétés" of Paris under N°844 425 629. UBS Europe SE Succursale de France is subject to the joint supervision of the European Central Bank, the German Central Bank (Deutsche Bundesbank), the German Federal Financial Services Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, "BaFin"), as well as of the French "Autorité de contrôle prudentiel et de résolution" and "Autorité des marchés financiers", to which this publication has not been submitted for approval. UBS Europe SE is a credit institution constituted under German law in the form of a Societas Europaea, duly authorized by BaFin. **Germany**: This publication is not intended to constitute a public offer under German law. It is distributed only for information purposes to clients of UBS Europe SE, Germany, with place of business at Bockenheimer Landstrasse 2-4, 60306 Frankfurt am Main. UBS Europe SE is a credit institution constituted under German law in the form of a Societas Europaea, duly authorized by the German Federal Financial Services Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, "BaFin") and supervised jointly by the European Central Bank, the German Central Bank (Deutsche Bundesbank) and BaFin, to which this publication has not been submitted for approval. **Hong Kong SAR**: This publication is distributed to clients of UBS AG Hong Kong Branch by UBS AG Hong Kong Branch, a licensed bank under the Hong Kong Banking Ordinance and a registered institution under the Securities and Futures Ordinance. UBS AG Hong Kong Branch is incorporated in Switzerland with limited liability. The contents of this material have not been reviewed by any regulatory authority in Hong Kong. **India**: UBS Securities India Private Ltd. (Corporate Identity Number U67120MH1996PTC097299) 2/F, 3 North Avenue, Maker Maxity, Bandra Kurla Complex, Bandra (East), Mumbai (India) 400051. Phone: +912261556000. It provides brokerage services bearing SEBI Registration Number: INZ000259830; Merchant Banking services bearing SEBI Registration Number: INM000013101 and Research Analyst services bearing SEBI Registration Number: INH000001204. Name of Compliance Officer Mr. Parameshwaran Shivaramakrishnan, Phone: +912261556151, Email: parameshwaran.s@ubs.com, Name of Grievance Officer Parameshwaran Shivaramakrishnan, Phone: +912261556151, Email: ol-ubs-sec-compliance@ubs.com. Registration granted by SEBI, and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. UBS AG, its affiliates or subsidiaries may have debt holdings or positions in the subject Indian company/companies. UBS AG, its affiliates or subsidiaries may have financial interests (e.g. loan/derivative products, rights to or interests in investments, etc.) in the subject Indian company/companies from time to time. Within the past 12 months, UBS AG, its affiliates or subsidiaries may have received compensation for non-investment banking securities-related services and/or non-securities services from the subject Indian company/companies. The subject company/companies may have been a client/clients of UBS AG, its affiliates or subsidiaries during the 12 months preceding the date of distribution of the research report with respect to investment banking and/or non-investment banking securities-related services and/or non-securities services. With regard to information on associates, please refer to the Annual Report at: https://www.ubs.com/global/en/about_ubs/investor_relations/annualreporting.html. The Research Annual Compliance report for UBS Securities India Private Limited is available on www.ubs.com/ubssi under Research tab. **Indonesia**: This communication and any offering material term sheet, research report, other product or service documentation or any other information (the "Material") sent with this communication was done so as a result of a request received by UBS from you and/or persons entitled to make the request on your behalf. Should you have received the Material erroneously, UBS asks that you kindly delete/destroy the communication and Material and inform UBS immediately. The Material, where provided, was provided for your information only and is not to be further distributed without the consent of UBS. None of the Material has been registered or filed under the prevailing laws and with any financial or regulatory authority in your jurisdiction. The Material may not have been approved, disapproved, endorsed, registered or filed with any financial or regulatory authority in your jurisdiction. UBS has not, by virtue of the Material, made available, issued any invitation to subscribe for or to purchase any investment (including securities or products or futures contracts). The Material is neither an offer nor a solicitation to enter into any transaction or contract (including futures contracts) nor is it an offer to buy or to sell any securities or products. The relevant investments will be subject to restrictions and obligations on transfer as set forth in the Material, and by receiving the Material you undertake to comply fully with such restrictions and obligations. You should carefully study and ensure that you understand and exercise due care and discretion in considering your investment objective, risk appetite and personal circumstances against the risk of the investment. You are advised to seek independent professional advice in case of doubt. Any and all advice provided on and/or trades executed by UBS pursuant to the Material will only have been provided upon your specific request or executed upon your specific instructions, as the case may be, and may be deemed as such by UBS and you. **Israel**: UBS is a premier global financial firm offering wealth management, asset management and investment banking services from its headquarters in Switzerland and its operations in over 50 countries worldwide to individual, corporate and institutional investors. This publication is intended for information only and is not intended as an offer to buy or solicitation of an offer. Furthermore, this publication is not intended as an investment advice. Nothing contrary to the above, no action has been, or will be, taken in Israel that would permit an offering of the product(s) mentioned in this document or a distribution of this document to the public in Israel. In particular, this document has not been reviewed or approved by the Israeli Securities Authority. The product(s) mentioned in this document is/are being offered to a limited number of sophisticated investors who qualify as one of the investors listed in the first supplement to the Israeli Securities Law, 5728-1968. This document may not be reproduced or used for any other purpose, nor be furnished to any other person other than those to whom copies have been sent. Anyone who purchases the product(s) mentioned herein shall do so for its own benefit and for its own account and not with the aim or intention of distributing or offering the product(s) to other parties. Anyone who purchases the product(s) shall do so in accordance with its own understanding and discretion and after it has received any relevant financial, legal, business, tax or other advice or opinion required by it in connection with such purchase(s). The word "advice" and/or any of its equivalent terms shall be read and construed in conjunction with the definition of the term "investment marketing" as defined under the Israeli Regulation of Investment Advice, Investment Marketing and Portfolio Management Law. The Swiss laws and regulations require a number of mandatory disclosures to be made in independent financial research reports or recommendations. Pursuant to the Swiss Financial Market Infrastructure Act and the Financial Market Infrastructure Ordinance-FINMA, banks must disclose the percentage of voting rights they hold in companies being researched, if these holdings are equal to or exceed the statutory thresholds. In addition, the Directives on the Independence of Financial Research, issued by the Swiss Bankers Association, mandate a number of disclosures, including the disclosure of potential conflicts of interest, the participation within previous 12 months in any securities issues on behalf of the company being researched, as well as the fact that remuneration paid to the financial analysts is based generally upon the performance of (i) the new issues department or investment banking; or (ii) securities trading performance (including proprietary trading) or sales. **Italy**: This publication is not intended to constitute a public offer under Italian law. It is distributed only for information purposes to clients of UBS Europe SE, Succursale Italia, with place of business at Via del Vecchio Politecnico, 3-20121 Milano. UBS Europe SE, Succursale Italia is subject to the joint supervision of the European Central Bank, the German Central Bank (Deutsche Bundesbank), the German Federal Financial Services Supervisory Authority (Bundesanstalt für

Finanzdienstleistungsaufsicht, "BaFin"), as well as of the Bank of Italy (Banca d'Italia) and the Italian Financial Markets Supervisory Authority (CONSOB - Commissione Nazionale per le Società e la Borsa), to which this publication has not been submitted for approval. UBS Europe SE is a credit institution constituted under German law in the form of a Societas Europaea, duly authorized by BaFin. **Japan**: This report is solely distributed in Japan by UBS SuMi TRUST Wealth Management Co., Ltd, Financial Instruments Dealer, Director-General of Kanto Local Finance Bureau (Kinsho) No. 3233, a member of the Japan Securities Dealers Association, Financial Futures Association of Japan, Investment Management Association of Japan. UBS SuMi TRUST Wealth Management Co., Ltd will not distribute or forward this report outside Japan. **Jersey**: UBS AG, Jersey Branch, is regulated and authorized by the Jersey Financial Services Commission for the conduct of banking, funds and investment business. Where services are provided from outside Jersey, they will not be covered by the Jersey regulatory regime. UBS AG, Jersey Branch is a branch of UBS AG a public company limited by shares, incorporated in Switzerland whose registered offices are at Aeschenvorstadt 1, CH-4051 Basel and Bahnhofstrasse 45, CH 8001 Zurich. UBS AG, Jersey Branch's principal place of business is 1, IFC Jersey, St Helier, Jersey, JE2 3BX. **Luxembourg**: This publication is not intended to constitute a public offer under Luxembourg law. It is distributed only for information purposes to clients of UBS Europe SE, Luxembourg Branch ("UBS Luxembourg"), R.C.S. Luxembourg n° B209123, with registered office at 33A, Avenue J. F. Kennedy, L-1855 Luxembourg. UBS Europe SE is a credit institution constituted under German law in the form of a Societas Europaea (HRB n° 107046), with registered office at Bockenheimer Landstrasse 2-4, D-60306 Frankfurt am Main, Germany, duly authorized by the German Federal Financial Services Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, "BaFin") and subject to the joint prudential supervision of BaFin, the European Central Bank and the central bank of Germany (Deutsche Bundesbank). UBS Luxembourg is furthermore supervised by the Luxembourg prudential supervisory authority (Commission de Surveillance du Secteur Financier), in its role as host member state authority. This publication has not been submitted for approval to any public supervisory authority. **Malaysia**: This communication and any offering material term sheet, research report, other product or service documentation or any other information (the "Material") sent with this communication was done so as a result of a request received by UBS from you and/or persons entitled to make the request on your behalf. Should you have received the Material erroneously, UBS asks that you kindly delete/destroy the communication and Material and inform UBS immediately. The Material, where provided, was provided for your information only and is not to be further distributed in whole or in part in or into your jurisdiction without the consent of UBS. The Material may not have been reviewed, approved, disapproved, endorsed, registered or filed with any financial or regulatory authority in your jurisdiction. UBS has not, by virtue of the Material, made available, issued any invitation to subscribe for or to purchase any investment (including securities or derivatives products). The Material is neither an offer nor a solicitation to enter into any transaction or contract (including future contracts) nor is it an offer to buy or to sell any securities or derivatives products. The relevant investments will be subject to restrictions and obligations on transfer as set forth in the Material, and by receiving the Material you undertake to comply fully with such restrictions and obligations. You should carefully study and ensure that you understand and exercise due care and discretion in considering your investment objective, risk appetite and personal circumstances against the risk of the investment. You are advised to seek independent professional advice in case of doubt. Any and all advice provided on and/or trades executed by UBS pursuant to the Material will only have been provided upon your specific request or executed upon your specific instructions, as the case may be, and may be deemed as such by UBS and you. **Mexico**: This information is distributed by UBS Asesores México, S.A. de C.V. ("UBS Asesores"), an affiliate of UBS Switzerland AG, incorporated as a non-independent investment advisor under the Mexican regulation due to the relation with a Foreign Bank. UBS Asesores is registered under number 30060-001-(14115)-21/06/2016 and subject to the supervision of the Mexican Banking and Securities Commission ("CNBV") exclusively regarding the rendering of (i) portfolio management services, (ii) securities investment advisory services, analysis and issuance of individual investment recommendations, and (iii) anti-money laundering and terrorism financing matters. This UBS publication or any material related thereto is addressed only to Sophisticated or Institutional Investors located in Mexico. Research reports only reflect the views of the analysts responsible for the report. The compensation of the analyst(s) who prepared this report is determined exclusively by research management and senior management of any entity of UBS Group to which such analyst(s) render(s) services. **Monaco**: This document is not intended to constitute a public offering or a comparable solicitation under the Principality of Monaco laws, but might be made available for information purposes to clients of UBS (Monaco) S.A., a regulated bank having its registered office at 2 avenue de Grande Bretagne 98000 Monaco operating under a banking license granted by the "Autorité de Contrôle Prudentiel et de Résolution" (ACPR) and the Monegasque government which authorizes the provision of banking services in Monaco. UBS (Monaco) S.A. is also licensed by the "Commission de Contrôle des Activités Financières" (CCAF) to provide investment services in Monaco. The latter has not approved this publication. **Philippines:** This communication was done so as a result of a request received by UBS from you and/or persons entitled to make the request on your behalf. Should you have received the Material erroneously, UBS asks that you kindly delete/destroy the communication and Material and inform UBS immediately. The Material, where provided, was provided for your information only and is not to be further distributed in whole or in part in or into your jurisdiction without the consent of UBS. The Material may not have been reviewed, approved, disapproved, endorsed, registered or filed with any financial or regulatory authority in your jurisdiction. UBS has not, by virtue of the Material, made available, issued any invitation to subscribe for or to purchase any investment (including securities or derivatives products). The Material is neither an offer nor a solicitation to enter into any transaction or contract (including future contracts) nor is it an offer to buy or to sell any securities or derivatives products. The relevant investments will be subject to restrictions and obligations on transfer as set forth in the Material, and by receiving the Material you undertake to comply fully with such restrictions and obligations. You should carefully study and ensure that you understand and exercise due care and discretion in considering your investment objective, risk appetite and personal circumstances against the risk of the investment. You are advised to seek independent professional advice in case of doubt. Any and all advice provided on and/or trades executed by UBS pursuant to the Material will only have been provided upon your specific request or executed upon your specific instructions, as the case may be, and may be deemed as such by UBS and you. **Portugal**: UBS Switzerland AG is not licensed to conduct banking and financial activities in Portugal nor is UBS Switzerland AG supervised by the portuguese regulators (Bank of Portugal "Banco de Portugal" and Portuguese Securities Exchange Commission "Comissão do Mercado de Valores Mobiliários"). **Qatar**: UBS Qatar LLC is licensed by the Qatar Financial Centre Authority and authorized by the QFC Regulatory Authority, with QFC no. 01169, and has its registered office at Building No. 36, 4th Floor, Abdulla Bin Thani Street, Doha Design District 1, Msheireb Downtown Doha, State of Qatar. This material is strictly intended for Eligible Counterparties and/or Business Customers only as classified under the QFCRA's Customer and Investor Protection Rules 2019. No other person should act upon this material. **Russia**: This document or information contained therein is for information purposes only and constitutes neither a public nor a private offering, is not an invitation to make offers, to sell, exchange or otherwise transfer any financial instruments in the Russian Federation to or for the benefit of any Russian person or entity and does not constitute an advertisement or offering of securities in the Russian Federation within the meaning of Russian securities laws. The information contained herein is not an "individual investment recommendation" as defined in Federal Law of 22 April 1996 No 39-FZ "On Securities Market" (as amended) and the financial instruments and operations specified herein may not be suitable for your investment profile or your investment goals or expectations. The determination of whether or not such financial instruments and operations are in your interests or are suitable for your investment goals, investment horizon or the acceptable risk level is your responsibility. We assume no liability for any losses connected with making any such operations or investing into any such financial instruments and we do not recommend to use such information as the only source of information for making an investment decision. **Saudi Arabia**: This material is for marketing and information purposes by UBS Saudi Arabia only. It has been distributed by UBS Saudi Arabia or reviewed and approved by UBS Saudi Arabia if it was issued by UBS Switzerland AG, by UBS AG, or any of its subsidiaries or branches. UBS Switzerland AG is a public company limited by shares, incorporated in Switzerland with its registered offices at Bahnhofstrasse 45, CH-8001 Zurich. UBS Saudi Arabia (a subsidiary of UBS AG) is a Saudi closed joint stock company incorporated in the Kingdom of Saudi Arabia with a paid capital of 110,000,000 Saudi Riyals under commercial register number 1010257812 having its registered office at Laysen Valley, P.O. Box 75724, Riyadh 11588, Kingdom of Saudi Arabia. UBS Saudi Arabia is authorized and regulated by the Capital Market Authority to conduct securities business under license number 08113-37. This document is distributed only under such circumstances as may be permitted by applicable laws and regulations. The information and – if any - opinions contained in this document have been compiled or arrived at based upon information obtained from sources believed to be reliable and in good faith, but is not guaranteed as being accurate, nor is it a complete statement or summary of the securities, markets or developments referred to in the document. All such information and opinions are subject to changes without notice. Past performance of investments (whether simulated or actual) is not necessarily an indicator of future results. Forward looking data is based on estimates that are subject to changes. Should the currency of a financial product or service not match your reference currency, performance may rise or fall due to currency fluctuations. UBS Switzerland AG and / or other members of the UBS Group may have a position in and may make a purchase and/or sale of any of the securities or other financial instruments mentioned in this document While statements that constitute "forward-looking statements", including, but not limited to, statements relating to our future business development represent UBS judgments and future expectations concerning the development of our business, a number of risks, uncertainties and other important factors could cause actual developments and results to differ materially from our expectations. This document may not be forwarded, reproduced, redistributed or republished for any purpose without the written permission of UBS Saudi Arabia. Source for all data and charts (if not indicated otherwise) is UBS Switzerland AG. **Singapore**: Where applicable, this material is distributed in Singapore by UBS AG, Singapore Branch, which is licensed by the Monetary Authority of Singapore under the Banking Act 1970 to carry on banking business. UBS AG is incorporated in Switzerland with limited liability. UBS AG has a branch registered in Singapore (UEN S98FC5560C). This material has been prepared and issued for distribution in Singapore to institutional investors, accredited investors and expert investors (each as defined under the Financial Advisers Regulations (the "FAR")) only. By virtue of your status as an institutional investor, accredited investor, or expert investor, UBS AG is exempted from complying with certain requirements under the Financial Advisers Act 2001 (the "FAA"), the FAR and the relevant Notices and Guidelines issued thereunder, in respect of any financial advisory service which UBS AG may provide to you. These include exemptions from complying with: Section 34 of the FAA (pursuant to Regulation 33(1) of the FAR); Section 36 of the FAA (pursuant to Regulation 34(1) of the FAR); and Section 45 of the FAA (pursuant to Regulation 35(1) of the FAR). Singapore recipients and clients of UBS AG, Singapore Branch should contact UBS AG, Singapore Branch for any matters arising from, or in connection with, this material. Where applicable, this communication and any offering material term sheet, research report, other product or service documentation or any other information (the "Material") sent with this communication was done so as a result of a request received by UBS from you and/or persons entitled to make the request on your behalf. Should you have received the Material erroneously, UBS asks that you kindly delete/destroy the communication and Material and inform UBS immediately. The Material, where provided, was provided for your information only and is not to be further distributed in whole or in part in or into your jurisdiction without the consent of UBS. The Material may not have been reviewed, approved, disapproved or endorsed by any financial or regulatory authority in your jurisdiction. UBS has not, by virtue of the Material, made available, issued any invitation to subscribe for or to purchase any investment (including securities or products or futures contracts). The Material is neither an offer nor a solicitation to enter into any transaction or contract (including future contracts) nor is it an offer to buy or to sell any securities or products. The relevant investments will be subject to restrictions and obligations on transfer as set forth in the Material, and by receiving the Material you undertake to comply fully with such restrictions and obligations. You should carefully study and ensure that you understand and exercise due care and discretion in considering your investment objective, risk appetite and personal circumstances against the risk of the investment. You are

advised to seek independent professional advice in case of doubt. Any and all advice provided on and/or trades executed by UBS pursuant to the Material will only have been provided upon your specific request or executed upon your specific instructions, as the case may be, and may be deemed as such by UBS and you. **Spain:** This report is distributed in Spain by UBS AG, Sucursal en España, authorized under number 1460 in the Register by the Banco de España. **Sweden**: This publication is not intended to constitute a public offer under Swedish law. It is distributed only for information purposes to clients of UBS Europe SE, Sweden Bankfilial, with place of business at Regeringsgatan 38, 11153 Stockholm, Sweden, registered with the Swedish Companies Registration Office under Reg. No 516406-1011. UBS Europe SE, Sweden Bankfilial is subject to the joint supervision of the European Central Bank, the German Central Bank (Deutsche Bundesbank), the German Federal Financial Services Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, "BaFin"), as well as of the Swedish supervisory authority (Finansinspektionen), to which this publication has not been submitted for approval. UBS Europe SE is a credit institution constituted under German law in the form of a Societas Europaea, duly authorized by BaFin. **Taiwan**: This material is provided by UBS AG, Taipei Branch in accordance with laws of Taiwan, in agreement with or at the request of clients/prospects. **Thailand**: This communication and any offering material, term sheet, research report, other product or service documentation or any other information (the "Material") sent with this communication were done so as a result of a request received by UBS from you and/or persons entitled to make the request on your behalf. Should you have received the Material erroneously, UBS asks that you kindly delete/destroy the communication and Material and inform UBS immediately. The Material, where provided, was provided for your information only and is not to be further distributed in whole or in part in or into your jurisdiction without the consent of UBS. The Material may not have been reviewed, approved, disapproved, endorsed, registered or filed with any financial or regulatory authority in your jurisdiction. UBS has not, by virtue of the Material, made available, issued any invitation to subscribe for or to purchase any investment (including securities or derivatives products). The Material is neither an offer nor a solicitation to enter into any transaction or contract (including future contracts) nor is it an offer to buy or to sell any securities or derivatives products. The relevant investments will be subject to restrictions and obligations on transfer as set forth in the Material, and by receiving the Material you undertake to comply fully with such restrictions and obligations. You should carefully study and ensure that you understand and exercise due care and discretion in considering your investment objective, risk appetite and personal circumstances against the risk of the investment. You are advised to seek independent professional advice in case of doubt. Any and all advice provided and/or trades executed by UBS pursuant to the Material will only have been provided upon your specific request or executed upon your specific instructions, as the case may be, and may be deemed as such by UBS and you. **Türkiye**: The information in this document is not provided for the purpose of offering, marketing or sale of any capital market instrument or service in the Republic of Türkiye. Therefore, this document may not be considered as an offer made, or to be made, to residents of the Republic of Türkiye in the Republic of Türkiye. UBS Switzerland AG is not licensed by the Capital Markets Board of Türkiye (the CMB) under the provisions of the Capital Market Law (Law No. 6362). Accordingly, neither this document nor any other offering material related to the instrument/service may be utilized in connection with providing any capital market services to persons within the Republic of Türkiye without the prior approval of the CMB. However, according to article 15 (d) (ii) of the Decree No. 32 residents of the Republic of Türkiye are allowed to purchase or sell the financial instruments traded in financial markets outside of the Republic of Türkiye. Further to this, pursuant to article 9 of the Communiqué on Principles Regarding Investment Services, Activities and Ancillary Services No. III-37.1, investment services provided abroad to residents of the Republic of Türkiye based on their own initiative are not restricted. **United Arab Emirates (UAE) / DIFC / Abu Dhabi**: UBS is not licensed in the UAE by the Central Bank of the UAE nor by the Emirates' Securities and Commodities Authority and does not undertake banking activities in the UAE. This document is provided for your information only and does not constitute financial advice. **DIFC**: UBS AG Dubai Branch is regulated by the DFSA in the DIFC. This material is strictly intended for Professional Clients and/or Market Counterparties only as classified under the DFSA rulebook. It should not be distributed to Retail Clients. The Investment Research is provided for information purposes only and is not a recommendation or offer to buy/sell/hold a particular investment. The investment research may be out of date. You should seek investment advice before acting on the basis of the Investment Research. **Abu Dhabi**: UBS AG Abu Dhabi Branch is licensed and regulated by the Financial Services Regulatory Authority ("FSRA") of the Abu Dhabi Global Market. This material is intended solely for professional clients or market counterparties, as defined in the rules of the FSRA. It is not directed at, nor intended for, retail clients or any person who does not meet the criteria of a professional client or market counterparty. **United Kingdom**: This document is issued by UBS Wealth Management, a division of UBS AG which is authorised and regulated by the Financial Market Supervisory Authority in Switzerland. In the United Kingdom, UBS AG is authorised by the Prudential Regulation Authority and is subject to regulation by the Financial Conduct Authority and limited regulation by the Prudential Regulation Authority. Details about the extent of regulation by the Prudential Regulation Authority are available from us on request. A member of the London Stock Exchange. This publication is distributed to retail clients of UBS Wealth Management. **Ukraine**: UBS is a premier global financial services firm offering wealth management services to individual, corporate and institutional investors. UBS is established in Switzerland and operates under Swiss law and in over 50 countries and from all major financial centers. UBS is not registered and licensed as a bank/financial institution under Ukrainian legislation and does not provide banking and other financial services in Ukraine. UBS has not made, and will not make, any offer of the mentioned products to the public in Ukraine. No action has been taken to authorize an offer of the mentioned products to the public in Ukraine and the distribution of this document shall not constitute financial services for the purposes of the Law of Ukraine "On Financial Services and Financial Companies" dated 14 December 2021. Any offer of the mentioned products shall not constitute an investment advice, public offer, circulation, transfer, safekeeping, holding or custody of securities in the territory of Ukraine. Accordingly, nothing in this document or any other document, information or communication related to the mentioned products shall be interpreted as containing an offer, a public offer or invitation to offer or to a public offer, or solicitation of securities in the territory of Ukraine or investment advice under Ukrainian law. Electronic communication must not be considered as an offer to enter into an electronic agreement or other electronic instrument within the meaning of the Law of Ukraine "On Electronic Commerce" dated 3 September 2015. This document is strictly for private use by its holder and may not be passed on to third parties or otherwise publicly distributed. **USA**: Distributed to US persons only by UBS Financial Services Inc. or UBS Securities LLC, subsidiaries of UBS AG. UBS Switzerland AG, UBS Europe SE, UBS Bank, S.A., UBS BB Corretora de Câmbio, Títulos e Valores Mobiliários S.A., UBS Asesores México, S.A. de C.V., UBS SuMi TRUST Wealth Management Co., Ltd., UBS Wealth Management Israel Ltd. and UBS Menkul Degerler AS are affiliates of UBS AG. **UBS Financial Services Inc. accepts responsibility for the content of a report prepared by a non-US affiliate when it distributes reports to US persons. All transactions by a US person in the securities mentioned in this report should be effected through a US-registered broker dealer affiliated with UBS, and not through a non-US affiliate. The contents of this report have not been and will not be approved by any securities or investment authority in the United States or elsewhere. UBS Financial Services Inc. is not acting as a municipal advisor to any municipal entity or obligated person within the meaning of Section 15B of the Securities Exchange Act (the "Municipal Advisor Rule") and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of the Municipal Advisor Rule.** For information on the ways in which UBS Securities LLC manages conflicts and maintains independence of its UBS Global Research product; historical performance information; certain additional disclosures concerning UBS Global Research recommendations; and terms and conditions for certain third party data used in research report, please visit https://www.ubs.com/disclosures.

© UBS 2026. The key symbol and UBS are among the registered and unregistered trademarks of UBS. All rights reserved.

Credit Suisse Wealth Management Disclaimer

This disclaimer must be read in conjunction with "Risk information" and "Important Information About Sustainable Investing Strategies" sections of the Global Wealth Management Disclaimer above. You receive this document in your capacity as a client of Credit Suisse Wealth Management. Your personal data will be processed in accordance with the Credit Suisse privacy statement accessible at your domicile through the official Credit Suisse website https://www.credit-suisse.com. In order to provide you with marketing materials concerning our products and services, UBS Group AG and its subsidiaries may process your basic personal data (i.e. contact details such as name, e-mail address) until you notify us that you no longer wish to receive them. You can optout from receiving these materials at any time by informing your Relationship Manager.

Except as otherwise specified herein and/or depending on the local Credit Suisse entity from which you are receiving this report, this report is distributed by UBS Switzerland AG, authorised and regulated by the Swiss Financial Market Supervisory Authority (FINMA). **Saudi Arabia:** This document is being distributed by Credit Suisse Saudi Arabia l Part of UBS Group (CR Number 1010228645, NUN Number 7001515373), duly licensed and regulated by the Saudi Arabian Capital Market Authority pursuant to License Number 08104-37 dated 23/03/1429H corresponding to 21/03/2008AD. Credit Suisse Saudi Arabia's principal place of business is at King Khaled Road, Laysen Valley, Building number 6, 12329-2376, Riyadh, Saudi Arabia. Website: https://www.credit-suisse.com/sa/en/cssa.html.

© UBS 2026. The key symbol and UBS are among the registered and unregistered trademarks of UBS. All rights reserved.

UBS